

Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

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Supplementary Content

S1 Proof of Theorem 1 (DMOP-α Functional Forms)	S-2
S2 Proof of Theorem 2 (Optimization Convergence Analysis)	S-5
S3 Feynman-Kac Models and Monte Carlo Approximations	S-10
S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling	S-12
S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates	S-18
S6 Proof of Theorem 5 (Bias-Variance Analysis)	S-19
S7 Sharper Bias-Variance Analysis	S-20
S8 Figures for Bayesian Inference	S-50
S9 Justification for the Choice of Baseline Comparisons	S-53
S10 Algorithmic Parameters	S-54

S1 Proof of Theorem 1 (DMOP- α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1:n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i .

Lemma S1. Write $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta), \quad (\text{S1})$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider the case of MOP- α when $\alpha = 1$ and $\theta = \phi$. We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned} \hat{\mathcal{L}}^1(\theta) &:= \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} \\ &= \left(\frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}, \end{aligned} \quad (\text{S2})$$

where the third equality follows from the choice of $\alpha = 1$, and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{\nabla_{\theta} \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_{\theta} \left(\frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}}{\prod_{n=1}^N L_n^{\phi}} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S3})$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_{\theta} \hat{\ell}^1(\theta) := \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S4})$$

We proceed to decompose (S4). First, observe that as $\alpha = 1$,

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}} = \prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}}, \quad (\text{S5})$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the j -th prediction particle at timestep n . We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_{\theta} w_{n,j}^{P,\theta}}{w_{n,j}^{P,\theta}} = \nabla_{\theta} \log w_{n,j}^{P,\theta} = \nabla_{\theta} \log \left(\prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}} \right) \quad (\text{S6})$$

$$= \nabla_{\theta} \sum_{i=1}^n \left(\log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi} \right) \quad (\text{S7})$$

$$= \sum_{i=1}^n \nabla_{\theta} \log g_{i,j}^{A,P,\theta}. \quad (\text{S8})$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the j -th prediction particle up to timestep n ,

$$\nabla_{\theta} \sum_{n=1}^N \log g_{n,j}^{A,P,\theta} = \nabla_{\theta} \log \left(\prod_{n=1}^N g_{n,j}^{A,P,\theta} \right) \quad (\text{S9})$$

$$= \nabla_{\theta} \log \left(\prod_{n=1}^N f_{Y_n|X_n}(y_n^* | x_{n,j}^{A,P,\theta}; \theta) \right) \quad (\text{S10})$$

$$= \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S11})$$

Multiplying both sides of the expression by $w_{N,j}^{P,\theta}$ yields an expression for the gradient of the weights at timestep N ,

$$\nabla_{\theta} w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^N \nabla_{\theta} \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S12})$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\begin{aligned} \nabla_{\theta} \hat{\ell}^1(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta} \\ &= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \end{aligned} \quad (\text{S13})$$

Finally, observing that $\theta = \phi$ implies $w_{N,j}^{F,\theta} = 1$, we obtain

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta). \quad (\text{S14})$$

This yields the gradient estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) when applied to the bootstrap filter. $\square$

Note that the variance of the DMOP- α log-likelihood derivative estimate scales poorly with N when $\theta \neq \phi$. This can be seen by observing that

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \quad (\text{S15})$$

When $\theta \neq \phi$, we see that $w_{N,k_j}^{P,\theta} = O(c^N)$. When $\theta = \phi$, this is a special case of the Poyiadjis et al. (2011) estimator, which has $O(N^4)$ variance by a property of functionals of the particle filter (Lemma 2 of Karjalainen et al. (2023) or Proposition 11.3 of Chopin and Papaspiliopoulos (2020)).

Lemma S2. Write $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$ for the DMOP- α gradient estimate (or, equivalently, the gradient of MOP- α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S16})$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)} \quad (\text{S17})$$

as shorthand for the measurement density ratios. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \quad (\text{S18})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \quad (\text{S19})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)}. \quad (\text{S20})$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\nabla_{\theta} \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_{\theta} \log \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S21})$$

$$= \sum_{n=1}^N \frac{\nabla_{\theta} \left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S22})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_{\theta} s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S23})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_{\theta} f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \quad (\text{S24})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left(f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S25})$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^J s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. $\square$

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

The analysis in this section is similar to the approach of Roosta-Khorasani and Mahoney (2016), with the caveat that none of the matrix concentration bounds they use apply here as the particles are dependent. We instead use the concentration inequality from Del Moral and Rio (2011) to bound the gradient and Hessian estimates. In this section, we fix $\omega \in \Omega$ only within each filtering iteration, and analyze Algorithm 3 post-iterated filtering.

The convergence analysis in Theorem 2 is limited to the case where $-\ell$ is γ -strongly convex. Under regularity conditions, we expect the log-likelihood to be approximately quadratic in a neighborhood of its maximum, though in practice the likelihood surfaces for POMP models are often highly nonconvex globally. The convergence to an optimum, local or global, must therefore be sensitive to initialization.

For simplicity, we only provide the argument for DMOP-1 and note that the general case follows as the required ϵ -approximation can still be obtained for sufficiently large α . When $\alpha = 1$, the required J is given in the below lemmas. We start by bounding the error on the gradient estimate, in Lemma S3.

Lemma S3 (Concentration of Measure for Gradient Estimate). *Consider the gradient estimate obtained by MOP-1, which we know by Theorem 4 is consistent for the score, where $\theta = \phi$. For $\|\nabla_{\theta} \hat{\ell}^1(\theta) - \nabla_{\theta} \ell(\theta)\|_2$ to be bounded by ϵ with probability $1 - \delta$, we require*

$$J > \max \left\{ 2G(\theta) \frac{r_N \sqrt{p}}{\epsilon} \left(1 + h^{-1} \left(\log \left(\frac{2p}{\delta} \right) \right) \right), 8G(\theta)^2 \beta_N^2 \frac{p \log(2p/\delta)}{\epsilon^2} \right\}, \quad (\text{S26})$$

where $NG'(\theta) \leq G(\theta)$ are defined in Assumptions (A3) and (A4), $h(t) = \frac{1}{2}(t - \log(1+t))$, and β_N and r_N are two additional finite model-specific constants that do not depend on J , but do depend on N and p , as defined in Del Moral and Rio (2011). Equivalently, with probability at least $1 - \delta$, it holds that

$$\|\nabla_{\theta} \hat{\ell}^1(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq G(\theta) \left(\frac{r_N \sqrt{p}}{J} (1 + h^{-1}(\log(2p/\delta))) + \sqrt{\frac{2p \log(2p/\delta)}{J}} \beta_N \right). \quad (\text{S27})$$

Note that, according to Del Moral and Rio (2011), under suitable regularity conditions, r_N and β_N are linear in the trajectory length N . This corresponds to the finding by Poyiadjis et al. (2011) that the variance of the estimate is at least quadratic in the trajectory length, and their remark that the result of Del Moral and Doucet (2003) establishes that the L_p error is bounded by $O(N^2 J^{-1/2})$ (equivalently, the variance is bounded by $O(N^4 J^{-1})$) after accounting for the sum over timesteps. The MOP-1 variance upper bound is therefore in fact $O(N^4)$, in contrast to the MOP- α , where $\alpha < 1$, upper bound of $O(N)$.

Proof. We use the concentration inequality of Del Moral and Rio (2011) to bound the deviation of the gradient estimate from the gradient of the negative log-likelihood in the sup norm with a union bound. Fix $\theta = \phi$. From the decomposition in the proof of Lemma S1, as $w_{N,j}^{F,\theta} = 1$ when $\theta = \phi$, we have that

$$\nabla_{\theta} \hat{\ell}(\theta) := \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \sum_{n=1}^N w_{N,k_j}^{P,\theta} \nabla_{\theta} \log g_{n,k_j}^{A,\theta} = \frac{1}{J} \sum_{j=1}^J \sum_{n=1}^N \nabla_{\theta} \log g_{n,k_j}^{A,\theta}. \quad (\text{S28})$$

Define $\varphi_n^i(x_{n,j}^{F,\theta}) := \frac{\partial}{\partial \theta_i} \log g_{n,k_j}^{A,\theta}$, which is a functional of the filtering particles $x_{n,j}^{F,\theta} = x_{n,k_j}^{P,\theta}$. These are bounded measurable functionals bounded by $G'(\theta)$ by Assumption (A3). Therefore, these have bounded oscillation, satisfying the requirement that $\text{osc}\left(\frac{\partial}{\partial \theta_i} \varphi_i(x_{n,j}^{P,\theta})\right) \leq G'(\theta)$. Note that Del Moral and Rio (2011) in fact assume $\text{osc}(f) \leq 1$, so we simply scale their bound accordingly.

Now we apply the Hoeffding-type concentration inequality in Theorem 1.2 of Del Moral and Rio (2011), and a union bound over each $\varphi_n^i(x_{n,j}^{F,\theta})$, totaling N timesteps and p parameters, to find that

$$\max_{n=1,\dots,N} \left\| \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log g_{n,k_j}^{A,\theta} - \nabla_{\theta} \ell_n(\theta) \right\|_{\infty} \leq G'(\theta) \left(\frac{r_N}{J} (1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}} \beta_N \right) \quad (\text{S29})$$

with probability at least $1 - 2Np \exp(-t)$. The concentration inequality justifying (S29) directly concerns the error from the expectation under the filtering distribution. However, we replace this with $\nabla_{\theta} \ell_n(\theta)$ in (S29) since our regularity conditions ensure that the gradient of the measurement density is bounded and thus, by dominated convergence, the expectation of this gradient equals the gradient of its expectation. Summing over N , and noting that $NG'(\theta) \leq G(\theta)$, we therefore obtain, with the same probability, that

$$\left\| \frac{1}{J} \sum_{j=1}^J \sum_{n=1}^N \nabla_{\theta} \log g_{n,k_j}^{A,\theta} - \nabla_{\theta} \ell(\theta) \right\|_{\infty} \leq G'(\theta) N \left(\frac{r_N}{J} (1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}} \beta_N \right) \quad (\text{S30})$$

$$\leq G(\theta) \left(\frac{r_N}{J} (1 + h^{-1}(t)) + \sqrt{\frac{2t}{J}} \beta_N \right). \quad (\text{S31})$$

We split the δ failure probability among these $2Np$ terms, to find $\delta \leq 2Np \exp(-t)$, and therefore, $t \leq \log(2Np/\delta)$, where $h(t) = \frac{1}{2}(t - \log(1+t))$. The two additional model-specific parameters are β_t and r_t , which do not depend on J . The analogous bound for the 2-norm follows from scaling the right-hand side by $\sqrt{p}$, to require

$$\|\nabla_{\theta} \hat{\ell}(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq G(\theta) \left(\frac{r_N \sqrt{p}}{J} (1 + h^{-1}(\log(2p/\delta))) + \sqrt{\frac{2p \log(2p/\delta)}{J}} \beta_N \right). \quad (\text{S32})$$

We therefore need

$$J > \max \left\{ 2G(\theta) \frac{r_N \sqrt{p}}{\epsilon} \left(1 + h^{-1} \left(\log \left(\frac{2p}{\delta} \right) \right) \right), 8G(\theta)^2 \beta_N^2 \frac{p \log(2p/\delta)}{\epsilon^2} \right\}. \quad (\text{S33})$$

□

We proceed to study the particle Hessian estimate, providing a guarantee for its positive-definiteness in Lemma S4. This bound is relevant if one chooses to use a second-order method involving a particle Hessian estimate.

Lemma S4 (Minimum Eigenvalue Bound for Hessian Estimate). *Assume that the Hessian of the negative log-likelihood $H = \sum_{j=1}^J \mathbb{E}H_j$ has a minimum eigenvalue $0 < \gamma < 1$, and that $\mathbb{E}\lambda_{\min}(H_j) = \gamma' > 0$. If*

$$J > \max \left\{ \frac{2r_t(1 + h^{-1}(t)) + 2c}{\gamma'}, \frac{2(2t\beta_t^2 + c)^2}{\gamma'^2} \right\} \geq \frac{r_t(1 + h^{-1}(t))}{\gamma'} + \sqrt{2tJ}\beta_t/\gamma' + c/\gamma' \quad (\text{S34})$$

then $\hat{H}(\theta)$ is invertible and positive definite with minimum eigenvalue greater than or equal to $c \in (0, \sum_{j=1}^J \mathbb{E}\lambda_{\min}(H_j))$, with probability at least $1 - \exp(-t)$.

Proof. Write $\hat{H}(\theta) = \hat{H} = \sum_{j=1}^J H_j$ for the estimate of the negative of the Hessian, where H_j is an element of the outer sum over the J particles.

As the negative log-likelihood is convex, we want to bound the minimum eigenvalue of $\hat{H}(\theta)$ from below with high probability, so that all the eigenvalues of $\hat{H}(\theta)$ are positive with high probability. This ensures that the estimated Hessian is invertible and positive-definite.

It is known that the minimum eigenvalue of a symmetric matrix is concave. Therefore, it suffices to show that the first inequality in the below expression

$$0 < \sum_{j=1}^J \lambda_{\min}(H_j) \leq \lambda_{\min} \left(\sum_{j=1}^J H_j \right) = \lambda_{\min}(\hat{H}) \quad (\text{S35})$$

holds with high probability. We apply the particle Hoeffding concentration inequality from Del Moral and Rio (2011) to find that

$$\frac{1}{J} \sum_{j=1}^J \lambda_{\min}(H_j) - \mathbb{E}_{\tilde{\pi}_t} \lambda_{\min}(H_j) = \frac{1}{J} \sum_{j=1}^J \lambda_{\min}(H_j) - \gamma' \geq -\frac{r_t}{J}(1 + h^{-1}(t)) - \sqrt{\frac{2t}{J}}\beta_t \quad (\text{S36})$$

$$\sum_{j=1}^J \lambda_{\min}(H_j) \geq -r_t(1 + h^{-1}(t)) - \sqrt{2tJ}\beta_t + J\gamma', \quad (\text{S37})$$

with probability at least $1 - \exp(-t)$. Here, $h(t) = \frac{1}{2}(t - \log(1 + t))$. The two additional model-specific parameters are β_t and r_t , which do not depend on J .

We additionally require, for $c \in (0, \sum_{j=1}^J \mathbb{E}\lambda_{\min}(H_j))$,

$$\sum_{j=1}^J \lambda_{\min}(H_j) \geq -r_t(1 + h^{-1}(t)) - \sqrt{2tJ}\beta_t + J\gamma' \geq c, \quad (\text{S38})$$

$$J\gamma' \geq c + r_t(1 + h^{-1}(t)) + \sqrt{2tJ}\beta_t. \quad (\text{S39})$$

It is therefore sufficient to have

$$J > \max \left\{ \frac{2r_t(1 + h^{-1}(t)) + 2c}{\gamma'}, \frac{2(2t\beta_t^2 + c)^2}{\gamma'^2} \right\} \geq \frac{r_t(1 + h^{-1}(t))}{\gamma'} + \sqrt{2tJ}\beta_t/\gamma' + c/\gamma' \quad (\text{S40})$$

for $\hat{H}(\theta)$ to be invertible and positive definite with minimum eigenvalue greater than or equal to c with probability at least $1 - \exp(-t)$. $\square$

We now proceed with the proof of Theorem 2. In this analysis, we largely follow the proof of Theorem 6 in Roosta-Khorasani and Mahoney (2016). Define $\theta_\eta = \theta_m + \eta p_m$, where $p_m = -(H(\theta_m))^{-1}g(\theta_m)$. As in Roosta-Khorasani and Mahoney (2016), we want to show there is some iteration-independent $\tilde{\eta} > 0$ such that the Armijo condition

$$f(\theta_m + \eta p_m) \leq f(\theta_m) + \eta \beta p_m^T g(\theta_m), \quad (\text{S41})$$

holds for any $0 < \eta < \tilde{\eta}$ and some $\beta \in (0, 1)$. By an argument found in the beginning of the proof of Theorem 6 of Roosta-Khorasani and Mahoney (2016), we have that choosing J such that $\|\nabla_\theta \hat{\ell}(\theta_m) - \nabla_\theta \ell(\theta_m)\| \leq \epsilon$ and $\lambda_{\min}(H(\theta_m)) \geq c > 0$ for each m , yields

$$f(\theta_\eta) - f(\theta_m) \leq \eta p_m^T g(\theta_m) + \epsilon \eta \|p_m\| + \eta^2 \Gamma \|p_m\|^2 / 2, \quad (\text{S42})$$

with probability $1 - \delta/2$. From now on, we assume that we are on the success event of this high-probability statement. Consequently, we have

$$p_m^T g(\theta_m) = -p_m^T H(\theta_m) p_m \geq -c \|p_m\|^2, \quad (\text{S43})$$

and we can obtain a decrease in the objective. Substituting this into the previous expression,

$$f(\theta_\eta) - f(\theta_m) \leq -\eta p_m^T H(\theta_m) p_m + \epsilon \eta \|p_m\| + \eta^2 \Gamma \|p_m\|^2 / 2, \quad (\text{S44})$$

the Armijo condition becomes

$$-\eta p_m^T H(\theta_m) p_m + \epsilon \eta \|p_m\| + \eta^2 \Gamma \|p_m\|^2 / 2 \leq \eta \beta p_m^T g(\theta_m) = -\eta \beta p_m^T H(\theta_m) p_m \quad (\text{S45})$$

$$\epsilon \|p_m\| + \eta \Gamma \|p_m\|^2 / 2 \leq (1 - \beta) p_m^T H(\theta_m) p_m \quad (\text{S46})$$

$$\epsilon + \eta \Gamma \|p_m\| / 2 \leq c(1 - \beta) \|p_m\|. \quad (\text{S47})$$

This holds and guarantees an iteration-independent lower bound if

$$\eta \leq \frac{c(1 - \beta)}{\Gamma}, \quad \epsilon \leq \frac{c(1 - \beta)}{2\Gamma} \|g(\theta_m)\| \leq \frac{c(1 - \beta)}{2} \|p_m\|, \quad (\text{S48})$$

which is given by our choice of η . Now, first note that

$$\|g(\theta_m)\| - \|\nabla_\theta f(\theta_m)\| \leq \|g(\theta_m) - \nabla_\theta f(\theta_m)\| \leq \epsilon \implies \|\nabla_\theta f(\theta_m)\| \geq \|g(\theta_m)\| - \epsilon \quad (\text{S49})$$

and

$$\|\nabla_\theta f(\theta_m)\| - \|g(\theta_m)\| \leq \|\nabla_\theta f(\theta_m) - g(\theta_m)\| \leq \epsilon \implies \|g(\theta_m)\| \geq \|\nabla_\theta f(\theta_m)\| - \epsilon. \quad (\text{S50})$$

There are now two cases. If the algorithm terminates and $\|g(\theta_m)\| \leq \sigma\epsilon$, we can derive

$$\|\nabla_\theta f(\theta_m)\| \leq \|g(\theta_m)\| + \epsilon = \sigma\epsilon + \epsilon = (\sigma + 1)\epsilon. \quad (\text{S51})$$

If the algorithm does not terminate, then $\|g(\theta_m)\| > \sigma\epsilon$. Notice that

$$\epsilon \geq \|g(\theta_m) - \nabla_\theta f(\theta_m)\| \geq \|g(\theta_m)\| - \|\nabla_\theta f(\theta_m)\| \quad (\text{S52})$$

$$\|\nabla_{\theta} f(\theta_m)\| + \epsilon \geq \|g(\theta_m)\| \geq \sigma\epsilon \quad (\text{S53})$$

$$\|\nabla_{\theta} f(\theta_m)\| \geq \sigma\epsilon - \epsilon = (\sigma - 1)\epsilon \quad (\text{S54})$$

$$\frac{\|\nabla_{\theta} f(\theta_m)\|}{\sigma - 1} \geq \epsilon, \quad (\text{S55})$$

and now

$$\|\nabla_{\theta} f(\theta_m)\| - \epsilon \geq \|\nabla_{\theta} f(\theta_m)\| - \frac{\|\nabla_{\theta} f(\theta_m)\|}{\sigma - 1} \quad (\text{S56})$$

$$= \left(1 - \frac{1}{\sigma - 1}\right) \|\nabla_{\theta} f(\theta_m)\| \quad (\text{S57})$$

$$= \frac{\sigma - 2}{\sigma - 1} \|\nabla_{\theta} f(\theta_m)\| \quad (\text{S58})$$

$$\geq \frac{2}{3} \|\nabla_{\theta} f(\theta_m)\|. \quad (\text{S59})$$

Since $\|A^{-1}\| = 1/\sigma_{\min}(A)$,

$$p_m^T H(\theta_m) p_m = (- (H(\theta_m))^{-1} g(\theta_m))^T H(\theta_m) (- (H(\theta_m))^{-1} g(\theta_m)) \quad (\text{S60})$$

$$= g(\theta_m)^T (H(\theta_m))^{-1} g(\theta_m) \quad (\text{S61})$$

$$\geq \frac{1}{c} \|g(\theta_m)\|^2 \quad (\text{S62})$$

$$\geq \frac{1}{c} (\|\nabla_{\theta} f(\theta_m)\| - \epsilon)^2 \quad (\text{S63})$$

$$\geq \frac{4}{9c} \|\nabla_{\theta} f(\theta_m)\|^2. \quad (\text{S64})$$

From the assumption that f is γ -strongly convex, $\gamma I \preceq \nabla_{\theta}^2 - \ell \preceq \Gamma I$, by an implication of γ -strong convexity we have

$$f(\theta_m) - f(\theta^*) \leq \frac{1}{2\gamma} \|\nabla_{\theta} f(\theta_m)\|^2, \quad (\text{S65})$$

and we put together:

$$f(\theta_m) - f(\theta^*) \leq \frac{1}{2\gamma} \|\nabla_{\theta} f(\theta_m)\|^2 \leq \frac{9c}{4} \frac{1}{2\gamma} p_m^T H(\theta_m) p_m, \quad (\text{S66})$$

$$\frac{8\gamma}{9c} (f(\theta_m) - f(\theta^*)) \leq \frac{4}{9c} \|\nabla_{\theta} f(\theta_m)\|^2 \leq p_m^T H(\theta_m) p_m, \quad (\text{S67})$$

$$f(\theta_m) - f(\theta^*) \leq \frac{9c}{8\gamma} p_m^T H(\theta_m) p_m, \quad (\text{S68})$$

$$-\frac{8\gamma}{9c} (f(\theta_m) - f(\theta^*)) \geq -\frac{4}{9c} \|\nabla_{\theta} f(\theta_m)\|^2 \geq -p_m^T H(\theta_m) p_m. \quad (\text{S69})$$

From earlier, as the Armijo condition is fulfilled with our choice of η and ϵ ,

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta p_m^T H(\theta_m) p_m + \epsilon \eta \|p_m\| + \eta^2 \Gamma \|p_m\|^2 / 2 \quad (\text{S70})$$

$$\leq -\eta \beta p_m^T H(\theta_m) p_m \quad (\text{S71})$$

$$\leq -\eta\beta \frac{8\gamma}{9c} (f(\theta_m) - f(\theta^*)). \quad (\text{S72})$$

Therefore,

$$f(\theta_{m+1}) - f(\theta^*) = f(\theta_{m+1}) - f(\theta_m) + f(\theta_m) - f(\theta^*) \quad (\text{S73})$$

$$\leq f(\theta_m) - f(\theta^*) - \eta\beta \frac{8\gamma}{9c} (f(\theta_m) - f(\theta^*)) \quad (\text{S74})$$

$$= \left(1 - \eta\beta \frac{8\gamma}{9c}\right) (f(\theta_m) - f(\theta^*)). \quad (\text{S75})$$

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$ be sequences of probability measures on the state space $\mathcal{X}$. This is the sequence of prediction distributions $f_{X_n|Y_{1:n-1}}$, filtering distributions $f_{X_n|Y_{1:n}}$, and posterior distributions $f_{X_{1:n}|Y_{1:n}}$ that we seek to approximate with the particle filter. For any measurable bounded functional h , we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \rightarrow [0, 1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x , and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x . If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A .

Markov kernels and measures: For any measure η , any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h d\eta = \int h(x)\eta(dx), \quad (\text{S76})$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \quad (\text{S77})$$

$$(Mh)(x) = \int M(x, dy)h(y). \quad (\text{S78})$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \quad (\text{S79})$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\text{TV}} = \sup_{\|h\|_{\infty} \leq 1/2} |\mu(\phi) - \nu(\phi)| = \sup_{\text{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \quad (\text{S80})$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\text{TV}}(M) = \sup_{x, y \in \mathcal{X}} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}. \quad (\text{S81})$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \rightarrow [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}(y_n^*|x_{n,j}^F; \theta) = G_n(x_{n,j}^F)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*|\cdot)$ is the conditional density of the observed measurement at time n , where we condition on the filtering particle $x_{n,j}^F$ as an element of the state space.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0 , a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0} , the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \quad (\text{S82})$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n)M_{n+1}. \quad (\text{S83})$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ \dots \circ \Phi_{k+1}. \quad (\text{S84})$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P}, \quad (\text{S85})$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^F}. \quad (\text{S86})$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = (\Phi_n(\eta_n^J))^{\otimes J} = \left(\Phi_n \left(\frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P} \right) \right)^{\otimes J}, \quad (\text{S87})$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+k} \circ \dots \circ \mathbf{M}_n. \quad (\text{S88})$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on $\mathcal{X}^J$. This is because at every timestep n , we in fact draw $X_{n,j}^P | \{X_{n-1,1:J}^P = x_{n-1,1:J}^P\} \sim \mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \eta_0^{\otimes J} \mathbf{M}_{0,n}$ for $j = 1, \dots, J$.

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP- α . That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \quad (\text{S89})$$

for some ϵ dependent on $\bar{G}, \underline{G}, \bar{M}, \underline{M}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP- α itself.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP- α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters). *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$, where we inductively define given some $x_{n-1,j}^F, w_{n-1,j}^F$ comprising the filtering measure approximation π_n^J :*

1. *The prediction particles at timestep n , $x_{n,j}^P$, are drawn from $X_{n,j}^P \sim \pi_n^J M_n$, and the prediction weights are given by $w_{n,j}^P = w_{n-1,j}^F$.*

2. The prediction measure at timestep n is given by

$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \quad (\text{S90})$$

3. The particles are resampled at every timestep n , yielding indices k_j according to some arbitrary probabilities $(p_{n,j})_{j=1}^J$.

4. The filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P g_{n,j}}{\sum_{j=1}^J w_{n,j}^P g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \quad (\text{S91})$$

5. The prediction weights at the next timestep are given by $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}$.

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting). A pair of random vectors (X, W) drawn from some measure g **targets** another measure π if, for any measurable and bounded functional h ,

$$E_g[h(X) \cdot W] = E_\pi[h(X)]. \quad (\text{S92})$$

A particle approximation is a triangular array of pairs of random vectors $(X_j^J, W_j^J), j = 1, 2, \dots, J$. The particle approximation **targets** π if, for any measurable and bounded functional h ,

$$\frac{\sum_{j=1}^J h(X_j^J) W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{\text{a.s.}} E_\pi[h(X)] \quad (\text{S93})$$

as $J \rightarrow \infty$.

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure). Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Now, let $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ be a multinomial sample with indices k_j drawn from $\{(\tilde{X}_j^J, U_j^J)\}$ where $(\tilde{X}_j^J, U_j^J)$ is represented, on average, proportional to $\pi_j^J J$ times. Write

$$(Y_j^J, V_j^J) = (\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J). \quad (\text{S94})$$

If the importance sampling weights U_j / π_j are bounded, then $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$ targets f_X .

Proof. Note that as the Y_j^J are a subsample from X_j^J , h can be a function of Y as well as it is one for X . We then expand

$$\frac{\sum_j h(Y_j^J) V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \quad (\text{S95})$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f(X)]. \quad (\text{S96})$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} 0. \quad (\text{S97})$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \xrightarrow{a.s.} 0 \quad (\text{S98})$$

since an application of this result with $h(x) = 1$ provides almost sure convergence of the denominator in (S97). Write $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\sum_j Z_j^J := \sum_j \left(g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J) \pi_j^J \right) \xrightarrow{a.s.} 0. \quad (\text{S99})$$

Because the functional h and importance sampling weights $u_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\mathbb{E}[(Z_j^J)^4] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent given the $\tilde{X}_j^J$ and U_j^J ,

$$\begin{aligned} \mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] &= J \mathbb{E} \left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] \\ &\quad + 3J(J-1) \left(\mathbb{E}[(Z_j^J)^2 | (\tilde{X}_j^J, U_j^J)_{j=1}^J] \right) \end{aligned} \quad (\text{S100})$$

$$\leq CJ^2, \quad (\text{S101})$$

for some $C > 0$. Taking expectations on both sides yields

$$\mathbb{E} \left[\left(\sum_j Z_j^J \right)^4 \right] \leq CJ^2 \quad (\text{S102})$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P} \left(\left| \frac{1}{J} \sum_{j=1}^J Z_j^J \right| > \epsilon \right) \leq \frac{1}{\epsilon^4 J^4} \mathbb{E} \left[\left(\sum_{j=1}^J Z_j^J \right)^4 \right] \leq \frac{C}{\epsilon^4 J^2}, \quad (\text{S103})$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\sum_j h(X_{k_j}^J) \frac{u_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(X_j^J) \frac{u_j^J}{\pi_j^J} \xrightarrow{a.s.} 0. \quad (\text{S104})$$

Similarly, we also have that

$$\sum_j \frac{u_j^J}{\pi_j^J} \pi_j^J - \sum_j \frac{u_{k_j}^J}{\pi_{k_j}^J} \pi_{k_j}^J \xrightarrow{a.s.} 0, \quad (\text{S105})$$

and the result is proved. $\square$

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals). *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $\tilde{Z}_j^J \sim f_{Z|X}(\cdot | \tilde{X}_j^J)$ where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z . Then, if the U_j^J are bounded, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z .*

Proof. We want to show that, for any measurable bounded h ,

$$\frac{\sum_j h(\tilde{Z}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}[\mathbb{E}_{f_{Z|X}}[h(Z)|X]]. \quad (\text{S106})$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S107})$$

Let $\bar{U}_j^J = \frac{J U_j^J}{\sum_j U_j^J}$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1} \sum_j h(\tilde{Z}_j^J) U_j^J}{J^{-1} \sum_j U_j^J} = \frac{J^{-1} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J}{J^{-1} \sum_j \bar{U}_j^J}. \quad (\text{S108})$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J, \quad (\text{S109})$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \right] = \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(\tilde{Z}_j^J) \bar{U}_j^J | \tilde{X}_j^J, \bar{U}_j^J] \right] \quad (\text{S110})$$

$$= \lim_{J \rightarrow \infty} \mathbb{E} \left[\frac{1}{J} \sum_j \mathbb{E} [h(Z) | X = \tilde{X}_j^J] \bar{U}_j^J \right]. \quad (\text{S111})$$

Applying (S107) with $g(x) = \mathbb{E}[h(Z)|X = x]$, the average on the right hand side converges almost surely to $\mathbb{E}\{\mathbb{E}[h(Z)|X]\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)]. \quad (\text{S112})$$

□

Lemma S7 (Particle Posteriors). *Suppose that $\{(X_j^J, U_j^J), j = 1, \dots, J\}$ targets f_X . Also suppose that $(X_j'^J, U_j'^J) = (X_j^J, U_j^J f_{Z|X}(z^*|X_j^J))$. Then, if $U_j^J f_{Z|X}(z^*|X_j^J)$ and $U_j^J f_{Z|X}(z^*|X_j^J)/f_Z(z^*)$ are bounded, $\{(X_j'^J, U_j'^J)\}$ targets $f_{X|Z}(\cdot|z^*)$.*

Proof. Again, we want to show that

$$\frac{\sum_j h(X_j^J) \cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}[h(X)|z^*]. \quad (\text{S113})$$

We already have that for any measurable bounded g ,

$$\frac{\sum_j g(X_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S114})$$

Consider the following:

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \times \left(\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \right)^{-1}. \quad (\text{S115})$$

We will apply Equation (S114) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[h(X) f_{Z|X}(z^*|X)], \quad (\text{S116})$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[f_{Z|X}(z^*|X)] = f_Z(z^*). \quad (\text{S117})$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}[h(X) f_{Z|X}(z^*|X)]}{f_Z(z^*)} = \mathbb{E}_{f_X} \left[\frac{h(X) f_{Z|X}(z^*|X)}{f_Z(z^*)} \right] \quad (\text{S118})$$

$$= \mathbb{E}_{f_X} \left[h(X) \frac{f_{X|Z}(X|z^*)}{f_X(X)} \right] = \mathbb{E}_{f_{X|Z}}[h(X)|z^*], \quad (\text{S119})$$

to give the desired result.

□

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution). *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^F, w_{n-1,j}^F)\}$ targets π_{n-1} . Then, Lemma S6 tells us that $\{(X_{n,j}^P, w_{n,j}^P)\}$ targets η_n . Lemma S7 tells us that $\{(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta)\}$ therefore targets π_n . Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^F, w_{n,j}^F) = (X_{n,k_j}^P, w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}), \quad (\text{S120})$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n . $\square$

Proposition S1 (MOP-1 Targets the Filtering Distribution). *When $\alpha = 1$ or $\phi = \theta$, MOP- α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α , the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*; \theta)$. Then, Lemma S6 tells us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. Lemma S7 tells us that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta} g_{n,j}^\theta)\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta} g_{n,k_j}^\theta / g_{n,k_j}^\phi), \quad (\text{S121})$$

with resampling weights proportional to $g_{n,j}^\phi$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$. $\square$

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting). *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1:n-1}^*; \theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \quad (\text{S122})$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}. \quad (\text{S123})$$

where L_n^ϕ is as defined in the various algorithms.

Here, (S122) is a direct consequence of our earlier result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$. To see (S123), we write the numerator of (S5) as

$$L_n^\phi \sum_{j=1}^J \left[\frac{g_{n,j}^\theta}{g_{n,j}^\phi} w_{n,j}^{P,\theta} \right] \frac{g_{n,j}^\phi}{L_n^\phi} = L_n^\phi \sum_{j=1}^J w_{n,j}^{FC,\theta} \frac{g_{n,j}^\phi}{L_n^\phi}. \quad (\text{S124})$$

Using Lemma S5, we resample according to probabilities $\frac{g_{n,j}^\phi}{L_n^\phi}$ to see this is properly estimated by

$$L_n^\phi \sum_{j=1}^J w_{n,j}^{F,\theta}, \quad (\text{S125})$$

from which we obtain (S123). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left(L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right). \quad (\text{S126})$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^J w_{0,j}^{P,\theta} = J$ on the denominator and $\sum_{j=1}^J w_{N,j}^{F,\theta}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution, $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta)$, which is not a-priori the gradient of the estimand $\nabla_\theta \eta_n(h_\theta)$. We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all J below.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals). *Let $h_\theta : \mathcal{X} \rightarrow \mathbb{R}$ be a measurable bounded functional of particles, where $\eta_n^J(h_\theta)$ has two continuous derivatives uniformly bounded over all J by H^* for almost every $\omega \in \Omega$ and $\theta \in \Theta$. If it holds that $\eta_n^J(h_\theta) \xrightarrow{a.s.} \eta(h_\theta) = h_\theta^*$, then we also have that $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*$.*

Proof. Fix $\omega \in \Omega$. The sequence $(\nabla_\theta \eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ is uniformly bounded over all J , by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of $\eta_n^J(h_\theta)(\omega)|_{\theta=\theta'}$ is also uniformly bounded over all J for almost every $\omega \in \Omega$ and every $\theta' \in \Theta$. A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ is uniformly equicontinuous over θ for

almost every $\omega \in \Omega$. Explicitly, for almost every $\omega \in \Omega$ and every $\epsilon > 0$, there exists some $\delta(\omega) > 0$ such that for every $\|\theta - \theta'\|_\infty < \delta$ and every $J \in \mathbb{N}$ we have that

$$\|\eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\|_\infty < \epsilon. \quad (\text{S127})$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$ converges pointwise for $\theta = \phi$ and almost every $\omega \in \Omega$, and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit $\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega)$. Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every $\omega \in \Omega$,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega). \quad (\text{S128})$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \rightarrow \eta_n(h_\theta) = h_\theta^* \quad (\text{S129})$$

for almost every $\omega \in \Omega$. We then have that for almost every $\omega \in \Omega$,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \quad (\text{S130})$$

as we wanted. $\square$

Theorem 4 is proved as a corollary of Theorem S2, where we apply $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$, $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$, and then use the continuous mapping theorem.

S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP- α . The earlier truncation argument is retained below in the source only for comparison. A differentiable particle includes its location and tangent, so the particle-system forgetting result applies to the score functionals carried by that particle. With the manuscript's convention that $\lesssim$ suppresses the particle-system mixing-time factor, this gives the N/J -times-an- α -factor variance bound in Theorem S5. Section S7 gives the complete proof, together with a historical direct bound that remains useful at $\alpha = 1$.

Theorem S5 (MSE and variance bounds for DMOP- α). *Suppose $\alpha \in [0, 1]$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. There are constants $C < \infty$ and $\varrho \in (0, 1)$, independent of N, J , and α , such that*

$$\begin{aligned} \mathbb{E} \left\| \nabla_\theta \ell(\theta) - \nabla_\theta \hat{\ell}^\alpha(\theta) \right\|_2^2 &\lesssim p G'(\theta)^2 \left[\frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 + \frac{1}{J^2} \left(\sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \right)^2 + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right], \end{aligned} \quad (\text{S131})$$

$$\text{Var} \left(\nabla_\theta \hat{\ell}^\alpha(\theta) \right) \lesssim p G'(\theta)^2 \min \left\{ N^2, \frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \right\}. \quad (\text{S132})$$

In addition, the historical direct argument gives

$$\mathbb{E} \left\| \nabla_{\theta} \ell(\theta) - \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\|_2^2 \leq CpG'(\theta)^2 \min \left[N^2, \frac{N^3}{J} + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right]. \quad (\text{S133})$$

For $0 \leq \alpha < 1$, Equation (S131) implies

$$\mathbb{E} \left\| \nabla_{\theta} \ell(\theta) - \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)^2} + N^2(1-\alpha)^2 + \frac{N^2}{J^2(1-\alpha)^2} \right\}. \quad (\text{S134})$$

Proof. Theorem S6, proved in Section S7, gives Equation (S131). Lemma S13 gives Equation (S132), and Proposition S2 gives Equation (S133). Finally, Equation (S275) gives Equation (S134). $\square$

S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S5. Write $\nabla_{\theta} \ell_n^{\alpha}(\theta)$ for the n th conditional score produced by the deterministic differentiated population recursion. At $\theta = \phi$, its $\alpha = 1$ version is the exact conditional score $\nabla_{\theta} \ell_n(\theta)$. A differentiated particle is the particle carried by the differentiated algorithm: it includes its location and the corresponding simulator and weight tangents. Thus the particle-state bounds and mixing statements below apply directly to the differentiable particles used by DMOP; there is no separate “physical particle” whose tangent must be treated afterward. Throughout this section, the observations are fixed. Expectations and variances are over the simulator and particle-filter random variables, conditional on those observations.

We use Assumptions (A3) and (A5) in the precise pathwise sense already required by the differentiated MOP construction. For fixed simulator random numbers and fixed resampling indices, Assumption (A5) says that the recursively simulated state path and the composed log measurement densities are continuously differentiable in θ near ϕ , and that the joint law of the base simulator random numbers does not depend on θ . The derivative in the last clause of Assumption (A3) is the resulting total derivative: uniformly in n, j and in a neighborhood of ϕ , after enlarging $G'(\theta)$ to bound that neighborhood if necessary,

$$\left\| \nabla_{\theta} \log g_{n,j}^{A,F,\theta} \right\|_{\infty} \leq G'(\theta). \quad (\text{S135})$$

These two existing assumptions also justify differentiation under the finite-horizon simulator expectations used below. Indeed, the product rule and Equation (S135) give, pathwise,

$$\begin{aligned} & \left\| \nabla_{\theta} \prod_{s=1}^t G_s(X_s) \right\|_{\infty} \\ &= \left\| \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_{\theta} \log G_s(X_s) \right\|_{\infty} \\ &\leq \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \left\| \nabla_{\theta} \log G_s(X_s) \right\|_{\infty} \\ &\leq t \bar{G}^t G'(\theta). \end{aligned} \quad (\text{S136})$$

The final expression is deterministic and finite for each fixed t . For each coordinate of θ , the mean-value theorem therefore bounds every difference quotient about ϕ by the same quantity $t\bar{G}^t G'(\theta)$. The dominated convergence theorem gives

$$\nabla_{\theta} \mathbb{E}_U \left[\prod_{s=1}^t G_s(X_s) \right] = \mathbb{E}_U \left[\nabla_{\theta} \prod_{s=1}^t G_s(X_s) \right]. \quad (\text{S137})$$

Thus the required interchange follows from the clarified Assumptions (A3) and (A5); it is not an additional assumption. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2), and the particle weak-error bound follows from the full-genealogy argument in Lemma S11.

For the path functionals below, evaluated at $\theta = \phi$, let ρ_t be the posterior path measure of the differentiable particle. Thus, for every integrable path functional h ,

$$\rho_t(h) = \frac{\mathbb{E} [h(X_{0:t}) \prod_{s=1}^t G_s(X_s)]}{\mathbb{E} [\prod_{s=1}^t G_s(X_s)]}. \quad (\text{S138})$$

Here X_s already includes its tangent, as stipulated above, while $G_s(X_s)$ is the existing measurement potential. Equation (S138) is therefore the usual posterior path measure on the state used by the differentiated particle filter, not an additional extension of it.

Unrolling the existing filtered-weight recursion at $\theta = \phi$ gives

$$\nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \nabla_{\theta} \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \quad (\text{S139})$$

For a complete ancestral path, write h_k , only in this section, for the actual vector mark

$$h_k = \nabla_{\theta} \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \quad (\text{S140})$$

carried on that path. In the path-space notation of Chan et al. (2018), $m(\xi_t)(h_k)$ is its empirical average among the extended ancestral paths at time t , and Equation (S138) gives its population counterpart.

We now identify the lag coefficients in the conditional score. At $\theta = \phi$, all weights equal one, so differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \end{aligned} \quad (\text{S141})$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^{\alpha}$. In the MOP- α recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time- n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \quad (\text{S142})$$

Consequently, the second average in Equation (S141) is pathwise

$$\frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} = \frac{\alpha}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \quad (\text{S143})$$

which is the empirical average over the time- $(n-1)$ filtered genealogy. Substituting Equation (S139) into Equation (S141) and changing the index in the second sum give

$$\begin{aligned} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}. \end{aligned} \quad (\text{S144})$$

Consequently, define

$$\begin{aligned} \Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \leq r < n. \end{aligned} \quad (\text{S145})$$

We now verify that the right-hand population quantities are the lag coefficients of the deterministic differentiated recursion. Equations (S137) and (S140) give, at $\theta = \phi$,

$$\begin{aligned} \nabla_{\theta} \log \mathbb{E}_U \left[\prod_{s=1}^n G_s(X_s) \right] &= \frac{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s) \sum_{s=1}^n h_s]}{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s)]} \\ &= \sum_{s=1}^n \rho_n(h_s). \end{aligned} \quad (\text{S146})$$

Subtracting the same identity at time $n-1$ gives

$$\begin{aligned} \nabla_{\theta} \ell_n(\theta) &= \sum_{s=1}^n \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\ &= \sum_{r=0}^{n-1} \Delta_{n,r}. \end{aligned} \quad (\text{S147})$$

Finally, differentiating the two population weighted averages in the deterministic MOP- α recursion and using Equation (S137) gives

$$\nabla_{\theta} \ell_n^{\alpha}(\theta) = \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r})$$

$$\begin{aligned}
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \{ \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \} \\
&= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}.
\end{aligned} \tag{S148}$$

Lemma S10 (Lagwise first-derivative forgetting). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0, 1)$ such that, for $0 \leq r < n$,*

$$\|\Delta_{n,r}\|_2 \leq C\sqrt{p} G'(\theta) \varrho^r. \tag{S149}$$

Proof. We derive the result from the posterior update and the contraction of the posterior backward kernels.

First suppose that $1 \leq r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of the mark in Equation (S140). By Equation (S135),

$$|h_{k,i}| \leq G'(\theta). \tag{S150}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \leq n - 1$, Equation (S138) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}
\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U \left[h_{k,i} \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{ G_n(X_n) \mid X_{n-1} \} \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{ G_n(X_n) \mid X_{n-1} \} \right]} \\
&= \frac{\rho_{n-1} \{ h_{k,i} M_n(G_n) \}}{\rho_{n-1} M_n(G_n)},
\end{aligned} \tag{S151}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n - 1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U \{ \prod_{s=1}^{n-1} G_s(X_s) \}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n). \tag{S152}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \geq \underline{G} \int M_n(x_{n-1}, dx_n) = \underline{G}, \tag{S153}$$

$$M_n(G_n)(x_{n-1}) \leq \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S154}$$

We next control how much information about the path through time k can remain at time $n - 1$. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1} , written only in this proof as B_s , is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S155}$$

Assumption (A2) gives the following minorization for every measurable set A :

$$\begin{aligned}
B_s(x_s, A) &= \frac{\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s | x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s | u; \theta)} \\
&\geq \frac{\underline{M} \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\
&= \frac{\underline{M}}{\bar{M}} \pi_{s-1}(A).
\end{aligned} \tag{S156}$$

Put $\epsilon = \underline{M}/\bar{M}$. If $\epsilon < 1$, Equation (S156) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1 - \epsilon) R_s(x_s, \cdot), \tag{S157}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1 - \epsilon} \tag{S158}$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x'_s ,

$$\begin{aligned}
B_s(f)(x_s) - B_s(f)(x'_s) &= (1 - \epsilon) \{R_s(f)(x_s) - R_s(f)(x'_s)\}, \\
|B_s(f)(x_s) - B_s(f)(x'_s)| &\leq (1 - \epsilon) \text{osc}(f).
\end{aligned} \tag{S159}$$

If $\epsilon = 1$, Equation (S156) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x'_s gives

$$\text{osc}\{B_s(f)\} \leq (1 - \epsilon) \text{osc}(f). \tag{S160}$$

The mark $h_{k,i}$ is a bounded function of the differentiable particle X_k . Repeated conditioning through the backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i}(X_k) \mid X_{n-1} = x_{n-1}\} = (B_{n-1}B_{n-2} \cdots B_{k+1}h_{k,i})(x_{n-1}). \tag{S161}$$

There are $n - 1 - k = r - 1$ backward kernels in this display. Moreover, Equation (S150) gives

$$\begin{aligned}
\text{osc}(h_{k,i}) &\leq 2\|h_{k,i}\|_\infty \\
&\leq 2G'(\theta).
\end{aligned} \tag{S162}$$

If $r = 1$, Equation (S161) contains no backward kernel. If $r \geq 2$, applying Equation (S160) successively gives

$$\begin{aligned}
&\text{osc}[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}] \\
&\leq 2G'(\theta) \begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2. \end{cases}
\end{aligned} \tag{S163}$$

To finish, abbreviate the conditional expectation on the left of Equation (S163) by H , only for the next calculation. Since the X_{n-1} -marginal of ρ_{n-1} is π_{n-1} , the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \tag{S164}$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \quad (\text{S165})$$

Substituting Equations (S164)– (S165) into Equation (S151) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned} \Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\ &= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)} \\ &= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}. \end{aligned} \quad (\text{S166})$$

For any probability measure μ and bounded real f ,

$$|f(x) - \mu(f)| \leq \text{osc}(f). \quad (\text{S167})$$

Equations (S153)– (S154) also imply

$$\text{osc}\{M_n(G_n)\} \leq \bar{G} - \underline{G}. \quad (\text{S168})$$

Taking absolute values in Equation (S166), using Equations (S167)– (S168), and then using Equation (S163), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\text{osc}(H) \text{osc}\{M_n(G_n)\}}{\underline{G}} \\ &\leq \frac{2(\bar{G} - \underline{G})}{\underline{G}} G'(\theta) \begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2. \end{cases} \end{aligned} \quad (\text{S169})$$

Choose any $\varrho \in (1 - \epsilon, 1)$. Since $r \geq 1$,

$$\begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2 \end{cases} \leq \varrho^{r-1} = \varrho^{-1} \varrho^r. \quad (\text{S170})$$

Absorbing the fixed factor ϱ^{-1} into C , summing the squared coordinate bounds, and taking square roots give

$$\begin{aligned} \|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^p |\Delta_{n,r,i}|^2 \right\}^{1/2} \\ &\leq C \sqrt{p} G'(\theta) \varrho^r, \quad 1 \leq r < n. \end{aligned} \quad (\text{S171})$$

Finally, when $r = 0$, Equation (S145) and the bounded-mark condition give

$$\begin{aligned} \|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\ &\leq \rho_n(\|h_n\|_2) \\ &\leq \sqrt{p} G'(\theta) \\ &\leq C \sqrt{p} G'(\theta) \varrho^0. \end{aligned} \quad (\text{S172})$$

Equations (S171) and (S172) prove Equation (S149).

□

Lemma S11 (Full-genealogy bias of a lag coefficient). *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is a constant independent of n, r, J such that, for $0 \leq r < n$,*

$$\|\mathbb{E}\Delta_{n,r}^J - \Delta_{n,r}\|_2 \leq \frac{C\sqrt{p}G'(\theta)}{J}n. \quad (\text{S173})$$

Proof. At $\theta = \phi$, resampling is multinomial with the ordinary bootstrap filter probabilities, so the differentiable particles together with their parent indices form the bootstrap particle genealogy. Fix $0 \leq r < n$, put $k = n - r$, and fix a parameter coordinate $i \in \{1, \dots, p\}$. For any ancestral path through a time $t \geq k$, let $h_{k,i}$ denote the i th coordinate of the existing mark h_k in Equation (S140); as a function on paths through time t , it ignores every coordinate after time k . Thus

$$m(\xi_t)(h_{k,i}) = \frac{1}{J} \sum_{j=1}^J h_{k,i}(\xi_t^j). \quad (\text{S174})$$

Its population counterpart is $\rho_t(h_{k,i})$ in Equation (S138). Since a tangent is a functional of the differentiable particle, the future normalizer below depends on a path prefix only through its last particle state. Equation (S135) implies, pointwise on every such path,

$$|h_{k,i}| \leq \|h_k\|_\infty \leq G'(\theta). \quad (\text{S175})$$

The Feynman–Kac convention of Chan et al. (2018) places in the target at horizon q the potentials through time $q - 1$. To represent the present filtered path law ρ_t , append at horizon $t + 1$ a selection with potential G_t , followed by the identity mutation. For this application only, put $G_0 = 1$, and, without loss of generality, decrease $\underline{G}$ and increase $\bar{G}$ so that $\underline{G} \leq 1 \leq \bar{G}$. This convention adds no observation at time zero and preserves Assumption (A3).

The resulting target at horizon $t + 1$ is ρ_t . To identify the particle measure, use for the terminal Chan–Del Moral–Jasra selection the same multinomial uniforms used in the manuscript’s time- t resampling. Both selections then choose the same parent path with probabilities proportional to G_t . The appended identity mutation adds no randomness; after deleting its repeated endpoint, its j th path is exactly ξ_t^j . Consequently, the terminal path-particle empirical measure is pathwise equal to $m(\xi_t)$, not merely equal to it in distribution.

We verify Assumption A1 of Chan et al. (2018) for this particle model. Let $0 \leq a \leq b \leq t + 1$, and let z and z' be two extended path prefixes whose last physical states are x_a and x'_a , respectively. As noted above, the future normalizer depends on each prefix only through that last physical state. If $a = b$, then

$$Q_{a,a}(1)(z) = Q_{a,a}(1)(z') = 1. \quad (\text{S176})$$

If $a < b$ and $a < t$, the Markov property and the definition of the Feynman–Kac semigroup give

$$Q_{a,b}(1)(z) = G_a(x_a) \int f_{X_{a+1}|X_a}(x_{a+1} | x_a; \theta) Q_{a+1,b}(1)(x_{a+1}) dx_{a+1}, \quad (\text{S177})$$

$$Q_{a,b}(1)(z') = G_a(x'_a) \int f_{X_{a+1}|X_a}(x_{a+1} | x'_a; \theta) Q_{a+1,b}(1)(x_{a+1}) dx_{a+1}. \quad (\text{S178})$$

The function $Q_{a+1,b}(1)$ is nonnegative. Assumptions (A2) and (A3) therefore give, term by term,

$$Q_{a,b}(1)(z) \leq \bar{G}\bar{M} \int Q_{a+1,b}(1)(x_{a+1}) dx_{a+1}, \quad (\text{S179})$$

$$Q_{a,b}(1)(z') \geq \underline{\text{GM}} \int Q_{a+1,b}(1)(x_{a+1}) dx_{a+1}. \quad (\text{S180})$$

The integral in these two displays is the same and is strictly positive. Consequently,

$$Q_{a,b}(1)(z) \leq \frac{\bar{M}\bar{G}}{\underline{\text{MG}}} Q_{a,b}(1)(z'). \quad (\text{S181})$$

For the remaining nonzero step, $a = t$ and $b = t + 1$, the appended identity mutation gives

$$\begin{aligned} Q_{t,t+1}(1)(z) &= G_t(x_t), & Q_{t,t+1}(1)(z') &= G_t(x'_t), \\ Q_{t,t+1}(1)(z) &\leq \frac{\bar{G}}{\underline{G}} Q_{t,t+1}(1)(z'). \end{aligned} \quad (\text{S182})$$

Together, Equations (S176) and (S181)–(S182) verify Assumption A1 of Chan et al. (2018), uniformly in a, b, t , with a constant no larger than $\bar{M}\bar{G}/(\underline{\text{MG}})$.

By Theorem 2 of Chan et al. (2018), after translating their particle count N to the present particle count J , there is a constant $C < \infty$, depending only on $\bar{M}, \underline{\text{M}}, \bar{G}, \underline{G}$, such that, whenever $J \geq C(t + 1)$, the asserted bound holds for nonnegative path functionals. Applying it to f^+ and f^- , and using $f = f^+ - f^-$, gives, for every integrable real path functional f ,

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq C \frac{t+1}{J} \rho_t(|f|). \quad (\text{S183})$$

Suppose first that $J \geq Cn$, with C enlarged so that the theorem's condition holds for $t \in \{n-1, n\}$. Since $t+1 \leq n+1 \leq 2n$ at these two times, substituting $f = h_{k,i}$ in Equation (S183), then using Equation (S175), gives

$$\begin{aligned} |\mathbb{E}m(\xi_t)(h_{k,i}) - \rho_t(h_{k,i})| &\leq C \frac{t+1}{J} \rho_t(|h_{k,i}|) \\ &\leq C \frac{t+1}{J} \rho_t\{G'(\theta)\} \\ &= \frac{CG'(\theta)(t+1)}{J} \\ &\leq \frac{CG'(\theta)n}{J}, \end{aligned} \quad (\text{S184})$$

When $r = 0$, we have $k = n$. The first line of Equation (S145) and Equation (S184) with $t = n$ yield

$$\begin{aligned} |\mathbb{E}\Delta_{n,0,i}^J - \Delta_{n,0,i}| &= |\mathbb{E}m(\xi_n)(h_{n,i}) - \rho_n(h_{n,i})| \\ &\leq \frac{CG'(\theta)n}{J}. \end{aligned} \quad (\text{S185})$$

When $1 \leq r < n$, we have $k = n - r \leq n - 1$, so the same path functional is defined at both times $n - 1$ and n . The second line of Equation (S145) gives

$$\begin{aligned} \mathbb{E}\Delta_{n,r,i}^J - \Delta_{n,r,i} &= \mathbb{E}\{m(\xi_n)(h_{k,i}) - m(\xi_{n-1})(h_{k,i})\} - \{\rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i})\} \\ &= \{\mathbb{E}m(\xi_n)(h_{k,i}) - \rho_n(h_{k,i})\} - \{\mathbb{E}m(\xi_{n-1})(h_{k,i}) - \rho_{n-1}(h_{k,i})\}. \end{aligned} \quad (\text{S186})$$

Taking absolute values, using $|a - b| \leq |a| + |b|$, and applying Equation (S184) at the two times gives

$$\begin{aligned}
|\mathbb{E}\Delta_{n,r,i}^J - \Delta_{n,r,i}| &\leq |\mathbb{E}m(\xi_n)(h_{k,i}) - \rho_n(h_{k,i})| + |\mathbb{E}m(\xi_{n-1})(h_{k,i}) - \rho_{n-1}(h_{k,i})| \\
&\leq \frac{CG'(\theta)n}{J} + \frac{CG'(\theta)(n-1)}{J} \\
&\leq \frac{2CG'(\theta)n}{J} \\
&\leq \frac{CG'(\theta)n}{J},
\end{aligned} \tag{S187}$$

where the final line again enlarges C . Combining Equations (S185) and (S187) give

$$\begin{aligned}
\|\mathbb{E}\Delta_{n,r}^J - \Delta_{n,r}\|_2 &\leq \left\{ \sum_{i=1}^p \left(\frac{CG'(\theta)n}{J} \right)^2 \right\}^{1/2} \\
&= \frac{C\sqrt{p}G'(\theta)n}{J}.
\end{aligned} \tag{S188}$$

It remains to remove the temporary restriction $J \geq Cn$. If $J < Cn$, Equation (S175) implies for every admissible t that

$$|m(\xi_t)(h_{k,i})| \leq m(\xi_t)(|h_{k,i}|) \leq G'(\theta), \tag{S189}$$

$$|\rho_t(h_{k,i})| \leq \rho_t(|h_{k,i}|) \leq G'(\theta). \tag{S190}$$

Thus, for $r = 0$,

$$|\mathbb{E}\Delta_{n,0,i}^J - \Delta_{n,0,i}| \leq 2G'(\theta), \tag{S191}$$

whereas, for $1 \leq r < n$, Equation (S186) gives

$$|\mathbb{E}\Delta_{n,r,i}^J - \Delta_{n,r,i}| \leq 4G'(\theta). \tag{S192}$$

The latter bound is valid in both cases. Therefore

$$\begin{aligned}
\|\mathbb{E}\Delta_{n,r}^J - \Delta_{n,r}\|_2 &\leq 4\sqrt{p}G'(\theta) \\
&< 4C\sqrt{p}G'(\theta)\frac{n}{J} \\
&\leq \frac{C\sqrt{p}G'(\theta)n}{J},
\end{aligned} \tag{S193}$$

after one final enlargement of C . This proves Equation (S173) for all $J \geq 1$. $\square$

Lemma S12 (Particle bias of the full DMOP- α score). *Suppose $\theta = \phi$. Under the conditions of Lemma S11, for every $\alpha \in [0, 1]$,*

$$\left\| \mathbb{E}\nabla_{\theta}\hat{\ell}^{\alpha}(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta) \right\|_2 \leq \frac{C\sqrt{p}G'(\theta)N^2}{J}, \tag{S194}$$

$$\left\| \mathbb{E}\nabla_{\theta}\hat{\ell}^{\alpha}(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta) \right\|_2 \lesssim \frac{\sqrt{p}G'(\theta)}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r. \tag{S195}$$

The constant depends only on $\bar{M}, \underline{M}, \bar{G}, \underline{G}$ and is independent of N, J , and α .

Proof. Sum Equation (S144) over $n = 1, \dots, N$ and put $k = n - r$ in the second sum. Since all sums are finite, their order may be interchanged, giving

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = \sum_{k=1}^N \left[m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \right]. \quad (\text{S196})$$

For fixed k , direct collection of the coefficient of every $m(\xi_t)(h_k)$ gives

$$\begin{aligned} & m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \\ &= (1 - \alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \end{aligned} \quad (\text{S197})$$

Indeed, the coefficient at $t = k$ is $1 - \alpha$; for $k < t < N$ it is $\alpha^{t-k} - \alpha^{t+1-k} = (1 - \alpha)\alpha^{t-k}$; and at $t = N$ it is α^{N-k} . Substitution of Equation (S197) into Equation (S196), followed by a second interchange of finite sums, yields

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} m(\xi_t) \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + m(\xi_N) \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S198})$$

The identical calculation, with $m(\xi_t)$ replaced by ρ_t in Equation (S148), gives

$$\nabla_{\theta} \ell^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} \rho_t \left(\sum_{k=1}^t \alpha^{t-k} h_k \right) + \rho_N \left(\sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S199})$$

The proof of Lemma S11, in particular Equation (S183), gives the following bound for every real bounded path functional f whenever $J \geq C(t + 1)$:

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq C \frac{t+1}{J} \|f\|_{\infty}. \quad (\text{S200})$$

If $J < C(t + 1)$, after taking $C \geq 1$, then

$$\begin{aligned} |\mathbb{E}m(\xi_t)(f) - \rho_t(f)| &\leq \mathbb{E}|m(\xi_t)(f)| + |\rho_t(f)| \\ &\leq 2\|f\|_{\infty} \\ &< 2C \frac{t+1}{J} \|f\|_{\infty}. \end{aligned} \quad (\text{S201})$$

Combining Equations (S200) and (S201), and enlarging C once, shows that for every $J \geq 1$,

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq C \frac{t+1}{J} \|f\|_{\infty}, \quad (\text{S202})$$

Fix a coordinate i . Equation (S135) gives

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \quad (\text{S203})$$

Subtracting Equation (S199) from Equation (S198), taking expectations, and then applying Equations (S202)–(S203) coordinatewise give

$$\begin{aligned}
& \left| \mathbb{E} \partial_{\theta_i} \hat{\ell}^\alpha(\theta) - \partial_{\theta_i} \ell^\alpha(\theta) \right| \\
& \leq \frac{CG'(\theta)}{J} \left[(1-\alpha) \sum_{t=1}^{N-1} (t+1) \sum_{k=1}^t \alpha^{t-k} + (N+1) \sum_{k=1}^N \alpha^{N-k} \right] \\
& = \frac{CG'(\theta)}{J} \left[\sum_{t=1}^{N-1} (t+1)(1-\alpha^t) + (N+1) \sum_{k=1}^N \alpha^{N-k} \right]. \tag{S204}
\end{aligned}$$

The last equality is valid at $\alpha = 1$ by direct evaluation of the product $(1-\alpha) \sum_{k=1}^t \alpha^{t-k}$. Since $1-\alpha^t \leq 1$ and $\sum_{k=1}^N \alpha^{N-k} \leq N$,

$$\begin{aligned}
& \sum_{t=1}^{N-1} (t+1)(1-\alpha^t) + (N+1) \sum_{k=1}^N \alpha^{N-k} \\
& \leq \sum_{t=1}^{N-1} (t+1) + N(N+1) \\
& = \frac{(N-1)(N+2)}{2} + N(N+1) \\
& = \frac{3N^2 + 3N - 2}{2} \leq 3N^2. \tag{S205}
\end{aligned}$$

Therefore

$$\left| \mathbb{E} \partial_{\theta_i} \hat{\ell}^\alpha(\theta) - \partial_{\theta_i} \ell^\alpha(\theta) \right| \leq \frac{CG'(\theta)N^2}{J}. \tag{S206}$$

Squaring this inequality, summing it over $i = 1, \dots, p$, and taking a square root proves Equation (S194).

It remains to prove the sharper α -dependent form. The time-uniform first-order weak-bias expansion for a bootstrap particle filter under strong mixing (Del Moral, 2004, Section 7.4.3) gives, for every bounded real function f of a current differentiable particle and every t ,

$$\left| \mathbb{E} \eta_t^J(f) - \eta_t(f) \right| \leq \frac{C}{J} \text{osc}(f). \tag{S207}$$

The uniformity in t can be seen directly in the cited first-order expansion. Assumption (A2) bounds the contribution of a one-step sampling error inserted l steps in the past by $C\varrho^l \text{osc}(f)/J$. Hence

$$\begin{aligned}
\sum_{l=0}^t \frac{C\varrho^l}{J} \text{osc}(f) & \leq \frac{C \text{osc}(f)}{J} \sum_{l=0}^{\infty} \varrho^l \\
& = \frac{C}{J(1-\varrho)} \text{osc}(f) \\
& \leq \frac{C}{J} \text{osc}(f), \tag{S208}
\end{aligned}$$

where the last line enlarges C by the fixed factor $(1-\varrho)^{-1}$. This proves the uniform $O(J^{-1})$ weak-bias estimate used here; no horizon-dependent path-space constant remains.

Equation (S139) and the bounded-mark condition give

$$\left\| \partial_{\theta_i} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} \right\|_{\infty} \leq G'(\theta) \sum_{r=0}^{n-1} \alpha^r, \quad \left\| \partial_{\theta_i} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \right\|_{\infty} \leq G'(\theta) \sum_{r=1}^{n-1} \alpha^r. \quad (\text{S209})$$

Apply Equation (S207) to these two weight-tangent coordinate functions in Equation (S141). Equation (S209), together with $\text{osc}(f) \leq 2\|f\|_{\infty}$, gives

$$\begin{aligned} & \left| \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) - \partial_{\theta_i} \ell_n^{\alpha}(\theta) \right| \\ & \leq \frac{CG'(\theta)}{J} \left(\sum_{r=0}^{n-1} \alpha^r + \sum_{r=1}^{n-1} \alpha^r \right) \\ & \leq \frac{CG'(\theta)}{J} \sum_{r=0}^{n-1} \alpha^r. \end{aligned} \quad (\text{S210})$$

Summing Equation (S210) over $n = 1, \dots, N$ gives

$$\begin{aligned} & \left| \mathbb{E} \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta) - \partial_{\theta_i} \ell^{\alpha}(\theta) \right| \\ & \leq \sum_{n=1}^N \left| \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) - \partial_{\theta_i} \ell_n^{\alpha}(\theta) \right| \\ & \leq \frac{CG'(\theta)}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r. \end{aligned} \quad (\text{S211})$$

Squaring Equation (S211), summing over the p coordinates, and taking square roots proves Equation (S195). Finally,

$$\begin{aligned} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r & \leq N \sum_{r=0}^{N-1} \alpha^r \\ & = N \frac{1 - \alpha^N}{1 - \alpha}, \end{aligned} \quad (\text{S212})$$

with the ratio interpreted as N at $\alpha = 1$. Thus Equation (S195) also implies Equation (S194) after the hidden constant is enlarged. $\square$

Lemma S13 (Historical-particle variance of the DMOP- α score). *Suppose $\theta = \phi$, $\alpha \in [0, 1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then*

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \leq CpG'(\theta)^2 \min \left\{ N^2, \frac{N^3}{J} \right\}, \quad (\text{S213})$$

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \lesssim pG'(\theta)^2 \min \left\{ N^2, \frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \right\}. \quad (\text{S214})$$

Here, as elsewhere in the supplement, $\lesssim$ suppresses the particle-system mixing-time factor. In particular,

$$\text{Var}\left(\nabla_{\theta}\hat{\ell}^{\alpha}(\theta)\right) \lesssim \frac{NpG'(\theta)^2}{J} \left(\frac{1-\alpha^N}{1-\alpha}\right)^2, \quad (\text{S215})$$

where the ratio is interpreted as N at $\alpha = 1$. The constant C is independent of N, J , and α , and $\text{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$.

Proof. We first prove the historical-particle estimate that will be applied. For $0 \leq a < b \leq t$, the one-step normalized kernel in the factorization used by Chan et al. (2018) assigns a measurable set A the probability

$$\frac{\int_A f_{X_{a+1}|X_a}(u | x_a; \theta) Q_{a+1,b}(1)(u) du}{\int f_{X_{a+1}|X_a}(u | x_a; \theta) Q_{a+1,b}(1)(u) du}. \quad (\text{S216})$$

The potential at time a cancels between the numerator and denominator. Assumption (A2) gives, for every x_a and A ,

$$\begin{aligned} & \frac{\int_A f_{X_{a+1}|X_a}(u | x_a; \theta) Q_{a+1,b}(1)(u) du}{\int f_{X_{a+1}|X_a}(u | x_a; \theta) Q_{a+1,b}(1)(u) du} \\ & \geq \frac{\underline{M} \int_A Q_{a+1,b}(1)(u) du}{\bar{M} \int Q_{a+1,b}(1)(u) du} \\ & = \frac{\underline{M}}{\bar{M}} \frac{\int_A Q_{a+1,b}(1)(u) du}{\int Q_{a+1,b}(1)(u) du}. \end{aligned} \quad (\text{S217})$$

The probability measure in the last line does not depend on x_a . Consequently, the Dobrushin coefficient of each kernel in Equation (S216) is at most

$$1 - \frac{\underline{M}}{\bar{M}}. \quad (\text{S218})$$

Writing $\eta_{a,b}^{x_a}$ for the corresponding conditional terminal law, as in Chan et al. (2018), composition of the $b - a$ kernels gives

$$\sup_{x_a, x'_a} \left\| \eta_{a,b}^{x_a} - \eta_{a,b}^{x'_a} \right\|_{\text{TV}} \leq 2 \left(1 - \frac{\underline{M}}{\bar{M}} \right)^{b-a}. \quad (\text{S219})$$

The appended identity mutation at horizon $t + 1$ can remove at most one contraction factor. Hence the summability condition in Assumption (A2) of Chan et al. (2018) follows from the explicit calculation

$$\sum_{d=0}^{\infty} 2 \left(1 - \frac{\underline{M}}{\bar{M}} \right)^{\max\{d-1, 0\}}$$

$$\begin{aligned}
&= 4 + 2 \sum_{d=2}^{\infty} \left(1 - \frac{\underline{M}}{\bar{M}}\right)^{d-1} \\
&= 4 + 2 \frac{1 - \underline{M}/\bar{M}}{\underline{M}/\bar{M}} \\
&= 2 + 2 \frac{\bar{M}}{\underline{M}} < \infty.
\end{aligned} \tag{S220}$$

It remains to verify the other hypotheses of that result. Let z and z' be two seed-state-tangent path prefixes through time a , with last physical states x_a and x'_a . Future potentials and physical-state transitions depend on these prefixes only through their last physical states, so, for $a \leq b \leq t+1$,

$$Q_{a,b}(1)(z) = Q_{a,b}(1)(x_a), \quad Q_{a,b}(1)(z') = Q_{a,b}(1)(x'_a). \tag{S221}$$

Equations (S176), (S181), and (S182) therefore imply

$$Q_{a,b}(1)(z) \leq \max \left\{ 1, \frac{\bar{M}\bar{G}}{\underline{M}\underline{G}} \right\} Q_{a,b}(1)(z'). \tag{S222}$$

Assumption (A3) gives, at every selection time,

$$0 < \underline{G} \leq G_a \leq \bar{G} < \infty. \tag{S223}$$

Thus Equations (S219)–(S223) verify Assumptions (A1)–(A2) of Chan et al. (2018) for the physical model. For $q \in \{1, 2\}$ and every $0 \leq F \leq 1$ on q extended paths, positivity and Equation (S221) give

$$\begin{aligned}
0 &\leq Q_{a,b}^{\otimes q}(F)(z_1, \dots, z_q) \\
&\leq Q_{a,b}^{\otimes q}(1)(z_1, \dots, z_q) \\
&= \prod_{l=1}^q Q_{a,b}(1)(x_a^l).
\end{aligned} \tag{S224}$$

The coalescence operators in Appendix C of Chan et al. (2018) are Markov kernels, and hence are positive and send 1 to 1. Equation (S224) therefore remains valid after every coalescence pattern, term by term. Together with Equation (S222), this shows that every bound in their Lemma C.1 and Proposition C.2 is unchanged by the seed and tangent marks; no total-variation contraction of those marks is invoked.

Apply the coalescent proof of Theorem 3.5 of Chan et al. (2018) to this marked historical model at the appended horizon $t+1$. Replacing their particle count by J and enlarging one constant $C \geq 2$ so that it applies to $q=1$ and $q=2$, for $0 \leq g \leq 1$ and $J > C(t+1)$ the theorem gives

$$\left(1 - C \frac{t+1}{J}\right) \rho_t(g) \leq \mathbb{E}g(\xi_t^1) \leq \left(1 + C \frac{t+1}{J}\right) \rho_t(g), \tag{S225}$$

$$\begin{aligned}
\left(1 - C \frac{t+1}{J}\right) \rho_t(g)^2 &\leq \mathbb{E}\{g(\xi_t^1)g(\xi_t^2)\} \\
&\leq \rho_t(g)^2 + C \frac{t+1}{J} \rho_t(g^2),
\end{aligned} \tag{S226}$$

where the terminal labels 1 and 2 are distinct. Since $0 \leq \rho_t(g), \rho_t(g^2) \leq 1$, the first line gives

$$|\mathbb{E}g(\xi_t^1) - \rho_t(g)| \leq C \frac{t+1}{J}, \quad (\text{S227})$$

and subtracting $\rho_t(g)^2$ in the second line gives

$$\begin{aligned} -C \frac{t+1}{J} \rho_t(g)^2 &\leq \mathbb{E}\{g(\xi_t^1)g(\xi_t^2)\} - \rho_t(g)^2 \\ &\leq C \frac{t+1}{J} \rho_t(g^2), \\ |\mathbb{E}\{g(\xi_t^1)g(\xi_t^2)\} - \rho_t(g)^2| &\leq C \frac{t+1}{J}. \end{aligned} \quad (\text{S228})$$

Let f be a bounded real path functional. If $\|f\|_\infty = 0$, the claim is immediate. Otherwise set

$$g = \frac{f + \|f\|_\infty}{2\|f\|_\infty}, \quad 0 \leq g \leq 1. \quad (\text{S229})$$

For $j \in \{1, 2\}$,

$$f(\xi_t^j) - \rho_t(f) = 2\|f\|_\infty \{g(\xi_t^j) - \rho_t(g)\}. \quad (\text{S230})$$

Exchangeability and Equations (S227)–(S228) therefore give

$$\begin{aligned} &|\mathbb{E} [\{f(\xi_t^1) - \rho_t(f)\} \{f(\xi_t^2) - \rho_t(f)\}]| \\ &= 4\|f\|_\infty^2 |\mathbb{E}\{g(\xi_t^1)g(\xi_t^2)\} - \rho_t(g)\mathbb{E}g(\xi_t^1) - \rho_t(g)\mathbb{E}g(\xi_t^2) + \rho_t(g)^2| \\ &\leq 4\|f\|_\infty^2 [|\mathbb{E}\{g(\xi_t^1)g(\xi_t^2)\} - \rho_t(g)^2| + \rho_t(g)|\mathbb{E}g(\xi_t^1) - \rho_t(g)| + \rho_t(g)|\mathbb{E}g(\xi_t^2) - \rho_t(g)|] \\ &\leq C \frac{t+1}{J} \|f\|_\infty^2. \end{aligned} \quad (\text{S231})$$

For $J > C(t+1)$, exchangeability of the J terminal paths gives

$$\begin{aligned} &\mathbb{E} |m(\xi_t)(f) - \rho_t(f)|^2 \\ &= \frac{1}{J^2} \sum_{j=1}^J \mathbb{E} |f(\xi_t^j) - \rho_t(f)|^2 \\ &\quad + \frac{1}{J^2} \sum_{\substack{1 \leq j, l \leq J \\ j \neq l}} \mathbb{E} [\{f(\xi_t^j) - \rho_t(f)\} \{f(\xi_t^l) - \rho_t(f)\}] \\ &= \frac{1}{J} \mathbb{E} |f(\xi_t^1) - \rho_t(f)|^2 + \frac{J-1}{J} \mathbb{E} [\{f(\xi_t^1) - \rho_t(f)\} \{f(\xi_t^2) - \rho_t(f)\}] \\ &\leq \frac{4}{J} \|f\|_\infty^2 + \frac{J-1}{J} |\mathbb{E} [\{f(\xi_t^1) - \rho_t(f)\} \{f(\xi_t^2) - \rho_t(f)\}]| \\ &\leq \frac{4}{J} \|f\|_\infty^2 + C \frac{t+1}{J} \|f\|_\infty^2 \\ &\leq C \frac{t+1}{J} \|f\|_\infty^2. \end{aligned} \quad (\text{S232})$$

For $J \leq C(t+1)$,

$$\mathbb{E} |m(\xi_t)(f) - \rho_t(f)|^2 \leq 4\|f\|_\infty^2$$

$$\leq 4C \frac{t+1}{J} \|f\|_\infty^2. \quad (\text{S233})$$

After enlarging C , the two cases prove, for every $J \geq 1$,

$$\mathbb{E} |m(\xi_t)(f) - \rho_t(f)|^2 \leq C \frac{t+1}{J} \|f\|_\infty^2. \quad (\text{S234})$$

Taking square roots gives

$$\left\{ \mathbb{E} |m(\xi_t)(f) - \rho_t(f)|^2 \right\}^{1/2} \leq C \sqrt{\frac{t+1}{J}} \|f\|_\infty. \quad (\text{S235})$$

Fix a coordinate i . Subtract Equation (S199) from Equation (S198), and use Minkowski's inequality in L^2 . Equations (S235) and (S203) then give

$$\begin{aligned} & \left\{ \mathbb{E} \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) - \partial_{\theta_i} \ell^\alpha(\theta) \right|^2 \right\}^{1/2} \\ & \leq (1-\alpha) \sum_{t=1}^{N-1} \left\{ \mathbb{E} \left| m(\xi_t) \left(\sum_{k=1}^t \alpha^{t-k} h_{k,i} \right) - \rho_t \left(\sum_{k=1}^t \alpha^{t-k} h_{k,i} \right) \right|^2 \right\}^{1/2} \\ & \quad + \left\{ \mathbb{E} \left| m(\xi_N) \left(\sum_{k=1}^N \alpha^{N-k} h_{k,i} \right) - \rho_N \left(\sum_{k=1}^N \alpha^{N-k} h_{k,i} \right) \right|^2 \right\}^{1/2} \\ & \leq \frac{CG'(\theta)}{\sqrt{J}} \left[(1-\alpha) \sum_{t=1}^{N-1} \sqrt{t+1} \sum_{k=1}^t \alpha^{t-k} + \sqrt{N+1} \sum_{k=1}^N \alpha^{N-k} \right] \\ & = \frac{CG'(\theta)}{\sqrt{J}} \left[\sum_{t=1}^{N-1} \sqrt{t+1} (1-\alpha^t) + \sqrt{N+1} \sum_{k=1}^N \alpha^{N-k} \right]. \end{aligned} \quad (\text{S236})$$

As in Equation (S204), the last equality is valid at $\alpha = 1$ by evaluating the product before using the geometric-sum formula. Since $1 - \alpha^t \leq 1$, $\sqrt{t+1} \leq \sqrt{N}$ for $t < N$, and the terminal geometric sum is at most N ,

$$\begin{aligned} & \sum_{t=1}^{N-1} \sqrt{t+1} (1-\alpha^t) + \sqrt{N+1} \sum_{k=1}^N \alpha^{N-k} \\ & \leq (N-1)\sqrt{N} + N\sqrt{N+1} \\ & \leq (1+\sqrt{2})N^{3/2}. \end{aligned} \quad (\text{S237})$$

Equations (S236)–(S237) imply

$$\mathbb{E} \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) - \partial_{\theta_i} \ell^\alpha(\theta) \right|^2 \leq \frac{CG'(\theta)^2 N^3}{J}. \quad (\text{S238})$$

For every square-integrable real random variable Y and deterministic a ,

$$\mathbb{E} |Y - a|^2 = \mathbb{E} |Y - \mathbb{E}Y|^2 + |\mathbb{E}Y - a|^2$$

$$\geq \text{Var}(Y). \quad (\text{S239})$$

Applying Equation (S239) to Equation (S238) and summing over the p coordinates gives

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \leq \frac{CpG'(\theta)^2 N^3}{J}. \quad (\text{S240})$$

We next use the fact that the weight tangent is part of the differentiable particle. Equation (S139) and Equation (S135) give, for every coordinate i ,

$$\begin{aligned} \left| \partial_{\theta_i} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} \right| &\leq \sum_{r=0}^{n-1} \alpha^r \left| \partial_{\theta_i} \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi} \right| \\ &\leq G'(\theta) \sum_{r=0}^{n-1} \alpha^r, \end{aligned} \quad (\text{S241})$$

$$\begin{aligned} \left| \partial_{\theta_i} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \right| &= \alpha \left| \partial_{\theta_i} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi} \right| \\ &\leq G'(\theta) \sum_{r=1}^{n-1} \alpha^r. \end{aligned} \quad (\text{S242})$$

The two terms in Equation (S141) are empirical averages of the bounded coordinate functions in Equations (S241) and (S242) of the differentiable particle state. The time-uniform L^4 particle bound in Lemma 2 of Karjalainen et al. (2023), applied to these two functions, gives

$$\left\| \frac{1}{J} \sum_{j=1}^J \partial_{\theta_i} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \mathbb{E} \left[\frac{1}{J} \sum_{j=1}^J \partial_{\theta_i} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} \right] \right\|_{L^4} \leq \frac{CG'(\theta)}{\sqrt{J}} \sum_{r=0}^{n-1} \alpha^r, \quad (\text{S243})$$

$$\left\| \frac{1}{J} \sum_{j=1}^J \partial_{\theta_i} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} - \mathbb{E} \left[\frac{1}{J} \sum_{j=1}^J \partial_{\theta_i} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \right] \right\|_{L^4} \leq \frac{CG'(\theta)}{\sqrt{J}} \sum_{r=1}^{n-1} \alpha^r. \quad (\text{S244})$$

Minkowski's inequality in L^4 , Equations (S141), (S243), and (S244) imply

$$\begin{aligned} &\left\| \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) - \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) \right\|_{L^4} \\ &\leq \frac{CG'(\theta)}{\sqrt{J}} \left(\sum_{r=0}^{n-1} \alpha^r + \sum_{r=1}^{n-1} \alpha^r \right) \\ &\leq \frac{CG'(\theta)}{\sqrt{J}} \sum_{r=0}^{n-1} \alpha^r. \end{aligned} \quad (\text{S245})$$

The conditional score at time n is a measurable function of the differentiable particle clouds at times $n-1$ and n . Theorem 3 of Karjalainen et al. (2023), together with the elementary fact that passing to a measurable functional can only reduce the generated σ -algebra and hence cannot increase a strong-mixing coefficient, therefore gives, for $l \geq 1$,

$$a(l) \leq (1 - \epsilon)^{\lfloor (l-1)/[c\{1+\log J\}] \rfloor}. \quad (\text{S246})$$

For $m < n$, Davydov's inequality, applied to the centered variables in Equation (S245), gives

$$\begin{aligned}
& \left| \text{Cov} \left(\partial_{\theta_i} \hat{\ell}_m^\alpha(\theta), \partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) \right) \right| \\
& \leq C a(n-m)^{1/2} \left\| \partial_{\theta_i} \hat{\ell}_m^\alpha(\theta) - \mathbb{E} \partial_{\theta_i} \hat{\ell}_m^\alpha(\theta) \right\|_{L^4} \left\| \partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) - \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) \right\|_{L^4} \\
& \leq \frac{C G'(\theta)^2}{J} a(n-m)^{1/2} \left(\sum_{r=0}^{m-1} \alpha^r \right) \left(\sum_{r=0}^{n-1} \alpha^r \right) \\
& \leq \frac{C G'(\theta)^2}{2J} a(n-m)^{1/2} \left\{ \left(\sum_{r=0}^{m-1} \alpha^r \right)^2 + \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \right\}, \tag{S247}
\end{aligned}$$

where the last line uses $2xy \leq x^2 + y^2$. Equation (S245) also gives

$$\text{Var} \left(\partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) \right) \leq \frac{C G'(\theta)^2}{J} \left(\sum_{r=0}^{n-1} \alpha^r \right)^2. \tag{S248}$$

Put $q = (1-\epsilon)^{1/2} < 1$. Directly summing the blocked geometric series in Equation (S246) gives

$$\begin{aligned}
\sum_{l=1}^{N-1} a(l)^{1/2} & \leq \sum_{l=1}^{N-1} q^{\lfloor (l-1)/\{c \log J\} \rfloor} \\
& \leq \min \left\{ N-1, \lceil c\{1 + \log J\} \rceil \sum_{u=0}^{\infty} q^u \right\} \\
& \leq C\{1 + \min(N, \log J)\}. \tag{S249}
\end{aligned}$$

Expanding the variance of the sum, using Equations (S247)–(S249), and observing that for each lag l every squared term occurs at most twice, we obtain

$$\begin{aligned}
& \text{Var} \left(\partial_{\theta_i} \hat{\ell}^\alpha(\theta) \right) \\
& = \sum_{n=1}^N \text{Var} \left(\partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) \right) + 2 \sum_{1 \leq m < n \leq N} \text{Cov} \left(\partial_{\theta_i} \hat{\ell}_m^\alpha(\theta), \partial_{\theta_i} \hat{\ell}_n^\alpha(\theta) \right) \\
& \leq \frac{C G'(\theta)^2}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 + \frac{C G'(\theta)^2}{J} \sum_{l=1}^{N-1} a(l)^{1/2} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \\
& \leq \frac{C G'(\theta)^2}{J} \{1 + \min(N, 1 + \log J)\} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2. \tag{S250}
\end{aligned}$$

Summing Equation (S250) over the p coordinates and then using the stated convention for $\lesssim$ gives the stochastic term in Equation (S214). Moreover,

$$\sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \leq N \left(\sum_{r=0}^{N-1} \alpha^r \right)^2$$

$$= N \left(\frac{1 - \alpha^N}{1 - \alpha} \right)^2, \quad (\text{S251})$$

with the last expression interpreted by continuity at $\alpha = 1$. Equations (S250) and (S251) prove Equation (S215).

It remains to obtain the other term in the minimum. Taking absolute values in Equation (S198) and using Equation (S203) give, pathwise,

$$\begin{aligned} \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) \right| &\leq G'(\theta) \left[\sum_{t=1}^{N-1} (1 - \alpha^t) + \sum_{k=1}^N \alpha^{N-k} \right] \\ &= G'(\theta) \left[(N-1) - \sum_{t=1}^{N-1} \alpha^t + 1 + \sum_{t=1}^{N-1} \alpha^t \right] \\ &= NG'(\theta). \end{aligned} \quad (\text{S252})$$

Consequently,

$$\text{Var} \left(\partial_{\theta_i} \hat{\ell}^\alpha(\theta) \right) \leq \mathbb{E} \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) \right|^2 \leq N^2 G'(\theta)^2. \quad (\text{S253})$$

Summing over coordinates and combining this bound with Equation (S240) proves Equation (S213). Combining the same deterministic bound with Equation (S250), summing over coordinates, and applying the $\lesssim$ convention proves Equation (S214). $\square$

Remark 1 (DMOP-0 and the fixed-lag variance calculation). *The earlier DMOP-0 and DMOP- (α, k) calculations can be retained, but the covariance premise must be stated explicitly. For $0 \leq k \leq N-1$, define the fixed-lag score by truncating the existing lag decomposition:*

$$\partial_{\theta_i} \hat{s}_k^\alpha(\theta) = \sum_{n=1}^N \sum_{r=0}^{\min\{k, n-1\}} \alpha^r \Delta_{n,r,i}^J. \quad (\text{S254})$$

Thus $\hat{s}_0^\alpha = \hat{\ell}^0$, independently of α , and $\hat{s}_{N-1}^\alpha = \hat{\ell}^\alpha$.

We first record the original DMOP-0 argument in its intended $\lesssim$ convention. If the strong-mixing estimate is understood to apply to the differentiated particle state, consisting of the physical particle together with its simulator tangent, Davydov's inequality and the centered fourth-moment particle bound give

$$|\text{Cov}(\Delta_{n,0,i}^J, \Delta_{m,0,i}^J)| \leq \frac{CG'(\theta)^2}{J} (1 - \epsilon)^{\frac{1}{2} \lfloor |n-m| / \{c \log J\} \rfloor}. \quad (\text{S255})$$

Therefore

$$\begin{aligned} \text{Var} \left(\partial_{\theta_i} \hat{\ell}^0(\theta) \right) &= \sum_{n,m=1}^N \text{Cov}(\Delta_{n,0,i}^J, \Delta_{m,0,i}^J) \\ &\leq \frac{CNG'(\theta)^2}{J} \{1 + \min(N, \log J)\}, \end{aligned} \quad (\text{S256})$$

after summing the blocked geometric series. Thus, with the manuscript's convention that $\lesssim$ suppresses this particle-system mixing-time factor,

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^0(\theta) \right) \lesssim \frac{NpG'(\theta)^2}{J}. \quad (\text{S257})$$

For the fixed-lag and full- α extensions, first suppose that there are $C < \infty$ and $\kappa \in (0, 1)$, independent of N, J, n, m, r , such that, for $0 \leq r \leq k$ and $r < n, m \leq N$,

$$|\text{Cov}(\Delta_{n,r,i}^J, \Delta_{m,r,i}^J)| \leq \frac{CG'(\theta)^2}{J} \kappa^{|n-m|}. \quad (\text{S258})$$

For each fixed r , summing the covariance matrix entries gives

$$\begin{aligned} & \text{Var} \left(\sum_{n=r+1}^N \Delta_{n,r,i}^J \right) \\ &= \sum_{n,m=r+1}^N \text{Cov}(\Delta_{n,r,i}^J, \Delta_{m,r,i}^J) \\ &\leq \frac{CG'(\theta)^2}{J} \sum_{n,m=r+1}^N \kappa^{|n-m|} \\ &\leq \frac{CG'(\theta)^2}{J} (N-r) \left(1 + 2 \sum_{l=1}^{\infty} \kappa^l \right) \\ &= \frac{CG'(\theta)^2}{J} (N-r) \frac{1+\kappa}{1-\kappa}. \end{aligned} \quad (\text{S259})$$

Centering Equation (S254), using Minkowski's inequality in L^2 , and then applying Equation (S259) yield

$$\begin{aligned} & \text{Var}(\partial_{\theta_i} \hat{s}_k^{\alpha}(\theta))^{1/2} \\ &\leq \sum_{r=0}^k \alpha^r \text{Var} \left(\sum_{n=r+1}^N \Delta_{n,r,i}^J \right)^{1/2} \\ &\leq \frac{CG'(\theta)}{\sqrt{J}} \sum_{r=0}^k \alpha^r \sqrt{N-r} \\ &\leq CG'(\theta) \sqrt{\frac{N}{J}} \sum_{r=0}^k \alpha^r. \end{aligned} \quad (\text{S260})$$

Squaring Equation (S260) and summing over the p coordinates give the finite, endpoint-continuous bound

$$\text{Var}(\nabla_{\theta} \hat{s}_k^{\alpha}(\theta)) \leq \frac{CNpG'(\theta)^2}{J} \left(\sum_{r=0}^k \alpha^r \right)^2$$

$$= \frac{CNpG'(\theta)^2}{J} \begin{cases} \{(1 - \alpha^{k+1})/(1 - \alpha)\}^2, & 0 \leq \alpha < 1, \\ (k+1)^2, & \alpha = 1. \end{cases} \quad (\text{S261})$$

In particular, $k = 0$ gives the non-polylogarithmic form of the earlier DMOP-0 conclusion whenever the uniform covariance bound in Equation (S258) is available:

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^0(\theta) \right) \leq \frac{CNpG'(\theta)^2}{J}. \quad (\text{S262})$$

For $k = N - 1$, Equation (S261) gives

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \leq \frac{CNpG'(\theta)^2}{J} \left(\frac{1 - \alpha^N}{1 - \alpha} \right)^2, \quad (\text{S263})$$

where the ratio is interpreted as N at $\alpha = 1$. Thus separate fixed-lag covariance control gives the requested N/J factor multiplied by a finite function of α , but for fixed $\alpha < 1$ that function has order $(1 - \alpha)^{-2}$.

The sharper power $(1 - \alpha)^{-1}$ requires covariance control jointly across the observation time and the injection time. Under Equation (S292) below, the two-dimensional convolution calculation in Equation (S293) instead gives

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \leq \frac{CpG'(\theta)^2}{J} \sum_{r=0}^{N-1} (N - r) \alpha^{2r}. \quad (\text{S264})$$

For $0 \leq \alpha < 1$, the finite sum is

$$\begin{aligned} \sum_{r=0}^{N-1} (N - r) \alpha^{2r} &= \frac{N - (N + 1)\alpha^2 + \alpha^{2N+2}}{(1 - \alpha^2)^2} \\ &\leq N \min \left\{ N, \frac{1}{1 - \alpha^2} \right\}. \end{aligned} \quad (\text{S265})$$

Its continuous value at $\alpha = 1$ is $N(N + 1)/2$. Hence Equation (S264) recovers $O(N/J)$ at $\alpha = 0$, gives $O\{N/[J(1 - \alpha)]\}$ for fixed $\alpha < 1$, and gives $O(N^2/J)$ at $\alpha = 1$.

The simulator tangent is a function of the differentiated particle state, so Equations (S255)–(S257) are exactly the original argument when the quoted strong-mixing estimate is understood for that augmented state. The bounded-density assumptions directly establish the quoted mixing theorem for the physical cloud; applying it to the augmented state–tangent cloud is the remaining scope condition, because the tangent is not generally recoverable from the physical locations alone. The $O(\log J)$ factor is not a separate objection under the manuscript’s $\lesssim$ convention. Truncating the discounted weights does not itself truncate the tangent, so the uniform fixed-lag and joint-lag covariance bounds are recorded separately. Equations (S261) and (S264) therefore restore the valid variance implications of the earlier argument, while Lemma S13 remains the unconditional result under the present assumptions.

Lemma S14 (Population discounting bias). *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0, 1]$. Then*

$$\|\nabla_{\theta} \ell_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n(\theta)\|_2 \leq C\sqrt{p}G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}. \quad (\text{S266})$$

Proof. Equation (S148) at α and at $\alpha = 1$ gives

$$\begin{aligned}\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} (1 - \alpha^r) \Delta_{n,r} \\ &= \sum_{r=1}^{n-1} (1 - \alpha^r) \Delta_{n,r},\end{aligned}\tag{S267}$$

because $1 - \alpha^0 = 0$. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}\|\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta)\|_2 &\leq \sum_{r=1}^{n-1} (1 - \alpha^r) \|\Delta_{n,r}\|_2 \\ &\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\ &\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{\infty} (1 - \alpha^r) \varrho^r \\ &= C\sqrt{p} G'(\theta) \left\{ \sum_{r=1}^{\infty} \varrho^r - \sum_{r=1}^{\infty} (\alpha\varrho)^r \right\} \\ &= C\sqrt{p} G'(\theta) \left\{ \frac{\varrho}{1 - \varrho} - \frac{\alpha\varrho}{1 - \alpha\varrho} \right\} \\ &= C\sqrt{p} G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}.\end{aligned}\tag{S268}$$

□

Theorem S6 (Refined MSE bound for DMOP- α). *Suppose the observations are fixed, $\theta = \phi$, $\alpha \in [0, 1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then there are $C < \infty$ and $\varrho \in (0, 1)$, independent of N, J , and α , such that*

$$\begin{aligned}\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ \lesssim p G'(\theta)^2 \left[\frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 + N^2 \left\{ \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)} \right\}^2 + \frac{1}{J^2} \left(\sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \right)^2 \right].\end{aligned}\tag{S269}$$

Consequently, for $0 \leq \alpha < 1$,

$$\begin{aligned}\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ \lesssim p G'(\theta)^2 \left\{ \frac{N}{J(1 - \alpha)^2} + N^2(1 - \alpha)^2 + \frac{N^2}{J^2(1 - \alpha)^2} \right\}.\end{aligned}\tag{S270}$$

Both displays are finite at $\alpha = 1$ when the finite sums in Equation (S269) are used; the three terms then have orders N^3/J , 0, and N^4/J^2 , respectively.

Proof. For a square-integrable random vector Z and a deterministic vector z , expansion of the square gives

$$\begin{aligned}\mathbb{E}\|Z - z\|_2^2 &= \mathbb{E}\|Z - \mathbb{E}Z + \mathbb{E}Z - z\|_2^2 \\ &= \mathbb{E}\|Z - \mathbb{E}Z\|_2^2 + 2\mathbb{E}\{(Z - \mathbb{E}Z)^\top(\mathbb{E}Z - z)\} + \|\mathbb{E}Z - z\|_2^2 \\ &= \text{Var}(Z) + \|\mathbb{E}Z - z\|_2^2,\end{aligned}\tag{S271}$$

because $\mathbb{E}(Z - \mathbb{E}Z) = 0$.

Apply Equation (S271) with $Z = \nabla_\theta \hat{\ell}^\alpha(\theta)$ and $z = \nabla_\theta \ell(\theta)$. Lemma S13 gives

$$\text{Var}\left(\nabla_\theta \hat{\ell}^\alpha(\theta)\right) \lesssim \frac{pG'(\theta)^2}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r\right)^2.\tag{S272}$$

For the bias, add and subtract the full population DMOP- α score. The triangle inequality, Lemma S12, and Lemma S14 give

$$\begin{aligned}&\left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2 \\ &\leq \left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2 \\ &\quad + \left\|\sum_{n=1}^N \{\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\}\right\|_2 \\ &\leq \frac{C\sqrt{p}G'(\theta)}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r + CN\sqrt{p}G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.\end{aligned}\tag{S273}$$

Using $(a+b)^2 \leq 2a^2 + 2b^2$ in Equation (S273) gives

$$\begin{aligned}&\left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\ &\leq CpG'(\theta)^2 \left[\frac{1}{J^2} \left(\sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r\right)^2 + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right].\end{aligned}\tag{S274}$$

Substitution of Equations (S272) and (S274) into Equation (S271) proves Equation (S269). For $\alpha < 1$,

$$\begin{aligned}\sum_{r=0}^{n-1} \alpha^r &\leq \frac{1}{1-\alpha}, \\ \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r\right)^2 &\leq \frac{N}{(1-\alpha)^2}, \\ \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r &\leq \frac{N}{1-\alpha}, \\ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} &\leq \frac{\varrho}{(1-\varrho)^2}(1-\alpha).\end{aligned}\tag{S275}$$

Substitution of Equation (S275) into Equation (S269), with the fixed factors in ϱ absorbed by $\lesssim$, proves Equation (S270). $\square$

Proposition S2 (Direct MSE companion bound). *Under the conditions of Theorem S6,*

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim p G'(\theta)^2 \left[\frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 + \frac{1}{J^2} \left(\sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \right)^2 + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right], \end{aligned} \quad (\text{S276})$$

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \leq C p G'(\theta)^2 \min \left[N^2, \frac{N^3}{J} + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right]. \end{aligned} \quad (\text{S277})$$

Thus Equation (S276) gives the sharper N/J -times-an- α -factor argument, while Equation (S277) retains the historical direct cap near $\alpha = 1$.

Proof. We first prove the sharper direct bound. Minkowski's inequality in L^2 gives

$$\begin{aligned} & \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \right\}^{1/2} \\ & \leq \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\|_2^2 \right\}^{1/2} \\ & \quad + \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2 \\ & \quad + \left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2. \end{aligned} \quad (\text{S278})$$

The three terms on the right of Equation (S278) are bounded by Lemma S13, Lemma S12, and Lemma S14, respectively. Hence

$$\begin{aligned} & \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \right\}^{1/2} \\ & \lesssim \sqrt{p} G'(\theta) \left[\left\{ \frac{1}{J} \sum_{n=1}^N \left(\sum_{r=0}^{n-1} \alpha^r \right)^2 \right\}^{1/2} + \frac{1}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r + N \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right]. \end{aligned} \quad (\text{S279})$$

For nonnegative a, b, c , $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$. Squaring Equation (S279) and applying this inequality proves Equation (S276).

For completeness, we next prove the historical direct cap. Summing Equation (S238) over the p coordinates gives

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 \leq \frac{C p G'(\theta)^2 N^3}{J}. \quad (\text{S280})$$

There is also a deterministic bound. Equation (S252) gives, for every coordinate i ,

$$\left| \partial_{\theta_i} \hat{\ell}^{\alpha}(\theta) \right| \leq N G'(\theta). \quad (\text{S281})$$

The identical calculation applied to Equation (S199), with $m(\xi_t)$ replaced by ρ_t , gives

$$|\partial_{\theta_i} \ell^\alpha(\theta)| \leq N G'(\theta). \quad (\text{S282})$$

The scalar triangle inequality, Equations (S281)–(S282), and then summation over coordinates give

$$\begin{aligned} & \left\| \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell^\alpha(\theta) \right\|_2^2 \\ &= \sum_{i=1}^p \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) - \partial_{\theta_i} \ell^\alpha(\theta) \right|^2 \\ &\leq \sum_{i=1}^p \left\{ \left| \partial_{\theta_i} \hat{\ell}^\alpha(\theta) \right| + \left| \partial_{\theta_i} \ell^\alpha(\theta) \right| \right\}^2 \\ &\leq 4pN^2 G'(\theta)^2. \end{aligned} \quad (\text{S283})$$

Combining Equations (S280) and (S283), and taking square roots, yields

$$\left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell^\alpha(\theta) \right\|_2^2 \right\}^{1/2} \leq C \sqrt{p} G'(\theta) \min \left\{ N, \frac{N^{3/2}}{\sqrt{J}} \right\}. \quad (\text{S284})$$

We next add the deterministic population discount error. The triangle inequality followed by Lemma S14 gives

$$\begin{aligned} & \left\| \nabla_{\theta} \ell^\alpha(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2 \\ &\leq \sum_{n=1}^N \left\| \nabla_{\theta} \ell_n^\alpha(\theta) - \nabla_{\theta} \ell_n(\theta) \right\|_2 \\ &\leq CN \sqrt{p} G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \end{aligned} \quad (\text{S285})$$

Minkowski's inequality in L^2 , followed by Equations (S284) and (S285), now gives

$$\begin{aligned} & \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \right\}^{1/2} \\ &\leq \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell^\alpha(\theta) \right\|_2^2 \right\}^{1/2} + \left\| \nabla_{\theta} \ell^\alpha(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2 \\ &\leq C \sqrt{p} G'(\theta) \left[\min \left\{ N, \frac{N^{3/2}}{\sqrt{J}} \right\} + N \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right]. \end{aligned} \quad (\text{S286})$$

Squaring Equation (S286) and using $(a+b)^2 \leq 2a^2 + 2b^2$ proves

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^\alpha(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ &\leq CpG'(\theta)^2 \left[\min \left\{ N^2, \frac{N^3}{J} \right\} + N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \right]. \end{aligned} \quad (\text{S287})$$

Finally, Equation (S281) at the given α and Equation (S282) at $\alpha = 1$ give, exactly as in Equation (S283),

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \leq 4pN^2 G'(\theta)^2. \quad (\text{S288})$$

Combining Equations (S287) and (S288), and enlarging C , proves Equation (S277). $\square$

Remark 2 (Relation to the direct historical bound of Del Moral and Doucet). *The proof of Proposition S2 follows the direct- L^2 strategy of Del Moral and Doucet (2003, Lemma 4), but uses the sharper two-lineage historical estimate already proved in Lemma S13. Applied literally to a norm-one bounded path functional, the generic inequality of Del Moral and Doucet (2003) has a sum of N uniformly bounded future-normalizer ratios and therefore gives RMSE $O(N/\sqrt{J})$. The unnormalized score has norm $O\{NG'(\theta)\}$, so homogeneity would give*

$$\text{RMSE} = O \left\{ \frac{N^2 G'(\theta)}{\sqrt{J}} \right\}, \quad \text{MSE} = O \left\{ \frac{N^4 G'(\theta)^2}{J} \right\}. \quad (\text{S289})$$

This is the generic historical rate recalled by Poyiadjis et al. (2011). The time-homogeneous specialization in Theorem 4 of Del Moral and Doucet (2003) uses their additional condition (23), which is not one of the present assumptions. By contrast, Equations (S227)–(S234) give RMSE $O(N^{3/2}/\sqrt{J})$ for the present bounded score functional and hence the sharper $O(N^3/J)$ direct MSE in Proposition S2. The direct proof removes the redundant N^4/J^2 term, but it does not remove the remaining historical factor N : that improvement still requires signed cross-time covariance control or stability of the inherited tangent.

Remark 3 (Why the generic historical bound is cubic). *The factor N^3/J in Theorem S6 is a generic upper bound, not a claim that the variance is always cubic. The historical occupation-measure theorem gives variance $O(N/J)$ for a norm-one path functional. At $\alpha = 1$, the terminal score is one path functional with norm $O(NG'(\theta))$; for general α , the summation-by-parts calculation gives the same uniform order. Homogeneity therefore gives*

$$\frac{N}{J} \{NG'(\theta)\}^2 = \frac{N^3 G'(\theta)^2}{J}. \quad (\text{S290})$$

Thus one factor N^2 comes from the squared scale of the unnormalized score and one factor N from the generic historical-particle variance constant. This is the same calculation carried out explicitly in Equations (S236)–(S240).

The population result in Lemma S10 does not remove the historical factor. It proves geometric decay of the deterministic coefficient $\Delta_{n,r}$, whereas the required variance calculation concerns centered finite-particle coefficients $\Delta_{n,r}^J$. The particle-cloud contraction of Karjalainen et al. (2023) cannot be substituted: it acts on the current physical cloud, not on the inherited simulator derivative, and its displayed blocks have length $O(\{\log J\})$.

The current-cloud covariance calculation elsewhere in this supplement therefore supplies a useful template but not the missing result. Even if the centered DMOP-0 increment were a function of the current physical cloud and its fourth centered moment had the usual particle order, Davydov's inequality and the displayed contraction of Karjalainen et al. (2023) would give, for $J \geq 2$,

$$\left| \text{Cov} \left(\partial_{\theta_i} \hat{\ell}_m^0(\theta), \partial_{\theta_i} \hat{\ell}_n^0(\theta) \right) \right| \leq \frac{CG'(\theta)^2}{J} (1 - \epsilon)^{\frac{1}{2} \lfloor |n-m|/\{c \log J\} \rfloor},$$

$$\sum_{m,n=1}^N \left| \text{Cov} \left(\partial_{\theta_i} \hat{\ell}_m^0(\theta), \partial_{\theta_i} \hat{\ell}_n^0(\theta) \right) \right| \leq \frac{CNG'(\theta)^2 \{1 + \min(N, \log J)\}}{J}. \quad (\text{S291})$$

Thus the covariance sum retains the particle-system mixing time; it is not $O(N/J)$ uniformly in J . For the actual DMOP mark there is a prior obstruction: the total fixed-random-number derivative carries the inherited simulator tangent and is not a function of the current physical cloud. Truncating the discounted weights does not truncate that inherited tangent. Hence neither the current-cloud calculation nor its truncated version proves geometric covariance for $\Delta_{n,r}^J$ under the present assumptions.

A genuinely sharper result would follow from a two-lineage covariance estimate for the existing lag coefficients. For example, if, for each coordinate i , one separately proved

$$\left| \text{Cov} \left(\Delta_{n,r,i}^J, \Delta_{m,s,i}^J \right) \right| \leq \frac{CG'(\theta)^2}{J} \kappa^{|n-m|} \lambda^{|(n-r)-(m-s)|}, \quad \kappa, \lambda \in (0, 1), \quad (\text{S292})$$

then, for $\alpha < 1$, substitution in Equation (S144) gives, coordinatewise,

$$\begin{aligned} & \text{Var} \left(\sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r,i}^J \right) \\ & \leq \frac{CG'(\theta)^2}{J} \sum_{n,m=1}^N \sum_{r=0}^{n-1} \sum_{s=0}^{m-1} \alpha^{r+s} \kappa^{|n-m|} \lambda^{|(n-r)-(m-s)|} \\ & \leq \frac{CG'(\theta)^2}{J} \frac{1+\kappa}{1-\kappa} \frac{1+\lambda}{1-\lambda} \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^{2r} \\ & = \frac{CG'(\theta)^2}{J} \frac{1+\kappa}{1-\kappa} \frac{1+\lambda}{1-\lambda} \sum_{r=0}^{N-1} (N-r) \alpha^{2r} \\ & \leq \frac{CG'(\theta)^2}{J} \frac{1+\kappa}{1-\kappa} \frac{1+\lambda}{1-\lambda} \frac{N}{1-\alpha^2} \\ & \leq \frac{CNG'(\theta)^2}{J(1-\alpha)}. \end{aligned} \quad (\text{S293})$$

The second inequality is the ℓ^2 convolution bound applied on the two coordinates n and $n-r$: the two convolution kernels have ℓ^1 norms $(1+\kappa)/(1-\kappa)$ and $(1+\lambda)/(1-\lambda)$, respectively. The equality counts the $N-r$ occurrences of lag r . The next inequality uses $\sum_{r=0}^{N-1} \alpha^{2r} \leq (1-\alpha^2)^{-1}$, and the last uses $1-\alpha^2 \geq 1-\alpha$. Summing Equation (S293) over the p coordinates would give the nonsingular finite-horizon bound

$$\text{Var} \left(\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \leq \frac{CpG'(\theta)^2}{J} \sum_{r=0}^{N-1} (N-r) \alpha^{2r} \quad (\text{S294})$$

$$\leq \frac{CpG'(\theta)^2}{J(1-\alpha)}, \quad \alpha < 1. \quad (\text{S295})$$

If one also proved the stronger geometrically lagged particle-bias estimate

$$\left\| \mathbb{E} \Delta_{n,r}^J - \Delta_{n,r} \right\|_2 \leq \frac{C\sqrt{p}G'(\theta)}{J} \varrho^r \quad (\text{S296})$$

uniformly in n, r, J , then Equations (S144) and (S148) would give

$$\begin{aligned}
& \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2 \\
& \leq \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \left\| \mathbb{E} \Delta_{n,r}^J - \Delta_{n,r} \right\|_2 \\
& \leq \frac{C \sqrt{p} G'(\theta)}{J} \sum_{n=1}^N \sum_{r=0}^{n-1} (\alpha \varrho)^r \\
& \leq \frac{CN \sqrt{p} G'(\theta)}{J(1 - \alpha \varrho)} \\
& \leq \frac{CN \sqrt{p} G'(\theta)}{J(1 - \varrho)}. \tag{S297}
\end{aligned}$$

Combining Equation (S297) with the population-bias bound in Lemma S14, and absorbing the fixed factor $(1 - \varrho)^{-2}$ into C , the MSE would satisfy

$$\text{MSE}_{\alpha} \leq CpG'(\theta)^2 \left\{ \frac{1}{J} \sum_{r=0}^{N-1} (N-r) \alpha^{2r} + N^2(1-\alpha)^2 + \frac{N^2}{J^2} \right\} \tag{S298}$$

$$\leq CpG'(\theta)^2 \left\{ \frac{N}{J(1-\alpha)} + N^2(1-\alpha)^2 + \frac{N^2}{J^2} \right\}, \quad \alpha < 1. \tag{S299}$$

Choosing $1 - \alpha \asymp (NJ)^{-1/3}$ in this conditional bound gives

$$\text{MSE}_{\alpha} \leq CpG'(\theta)^2 \left\{ \frac{N^{4/3}}{J^{2/3}} + \frac{N^2}{J^2} \right\}. \tag{S300}$$

For $1 < J < N^2$, both terms in Equation (S300) are smaller than N^2/J . Equations (S292)–(S300) are stated only to isolate the possible sharper regime. Neither the earlier SI arguments nor the current assumptions prove Equation (S292) for the reparameterized DMOP marks. Local physical-state marks make such a theorem plausible, but do not establish it; a proof still requires a two-lineage calculation. Alternatively, separate geometric stability of the augmented state–tangent recursion would address the missing dependence. Physical-state Doeblin mixing alone does not imply tangent stability.

Remark 4 (Ridge-type tuning and the endpoint derivatives). At $\theta = \phi$, the particle genealogy in Equation (S144) does not depend on α . Hence the actual MSE can be written, without any infinite series, as

$$\text{MSE}_{\alpha} = \mathbb{E} \left\| \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}^J - \nabla_{\theta} \ell(\theta) \right\|_2^2. \tag{S301}$$

It is a polynomial in α of degree at most $2(N-1)$, so the actual MSE has no singularity at $\alpha = 1$. Differentiating the finite sum in Equation (S301) gives

$$\frac{d}{d\alpha} \text{MSE}_{\alpha} = 2 \mathbb{E} \left[\left\{ \sum_{n=1}^N \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}^J - \nabla_{\theta} \ell(\theta) \right\}^{\top} \left\{ \sum_{n=1}^N \sum_{r=1}^{n-1} r \alpha^{r-1} \Delta_{n,r}^J \right\} \right]. \tag{S302}$$

In particular,

$$\left. \frac{d}{d\alpha} \text{MSE}_\alpha \right|_{\alpha=0+} = 2\mathbb{E} \left[\left\{ \sum_{n=1}^N \Delta_{n,0}^J - \nabla_\theta \ell(\theta) \right\}^\top \left\{ \sum_{n=2}^N \Delta_{n,1}^J \right\} \right], \quad (\text{S303})$$

$$\left. \frac{d}{d\alpha} \text{MSE}_\alpha \right|_{\alpha=1-} = 2\mathbb{E} \left[\left\{ \sum_{n=1}^N \sum_{r=0}^{n-1} \Delta_{n,r}^J - \nabla_\theta \ell(\theta) \right\}^\top \left\{ \sum_{n=1}^N \sum_{r=1}^{n-1} r \Delta_{n,r}^J \right\} \right]. \quad (\text{S304})$$

Thus moving right from DMOP-0 lowers the actual MSE if the right side of Equation (S303) is negative, and moving left from DMOP-1 lowers it if the right side of Equation (S304) is positive. If both inequalities are strict, continuity on $[0, 1]$ implies that a global MSE minimizer is interior and has smaller MSE than both endpoints. Neither sign follows from Assumptions (A2)–(A5), or from the absolute covariance bound in Equation (S292): lag contributions can be aligned or can cancel. This is the point at which the analogy with the classical ridge calculation requires additional signed covariance structure.

The proved bound in Theorem S6 cannot supply that structure. Its variance and particle-bias terms are independent of α , while direct differentiation of its population-bias factor gives

$$\begin{aligned} \frac{d}{d\alpha} N^2 \left\{ \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)} \right\}^2 \\ = -\frac{2N^2\varrho^2(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)^3} \leq 0. \end{aligned} \quad (\text{S305})$$

Consequently, the unconditional envelope is minimized at $\alpha = 1$ and does not establish an interior tradeoff.

There is, however, a rigorous ridge-type calculation for the conditional envelope in Equation (S298). After removing its common positive factor, the expression to minimize is

$$\frac{1}{J} \sum_{r=0}^{N-1} (N-r)\alpha^{2r} + N^2(1-\alpha)^2 + \frac{N^2}{J^2}. \quad (\text{S306})$$

For $N \geq 2$, the derivative of Equation (S306) at the two endpoints is

$$\left. \frac{d}{d\alpha} \right|_{\alpha=0} = -2N^2 < 0, \quad (\text{S307})$$

$$\begin{aligned} \left. \frac{d}{d\alpha} \right|_{\alpha=1} &= \frac{2}{J} \sum_{r=1}^{N-1} r(N-r) \\ &= \frac{2}{J} \left\{ N \frac{N(N-1)}{2} - \frac{N(N-1)(2N-1)}{6} \right\} \\ &= \frac{N(N^2-1)}{3J} > 0. \end{aligned} \quad (\text{S308})$$

Its second derivative is

$$\frac{2}{J} \sum_{r=1}^{N-1} r(2r-1)(N-r)\alpha^{2r-2} + 2N^2 > 0. \quad (\text{S309})$$

Equations (S307)–(S309) prove that this conditional envelope has a unique minimizer in $(0, 1)$ and that its value there is strictly smaller than its values at both endpoints. Retaining the finite sum is essential: at $\alpha = 1$ it equals $N(N + 1)/2$, so the apparent singularity in Equation (S299) is only an artifact of extending a finite geometric sum to infinity.

When $1 \ll J \ll N^2$, the minimizer is outside the width- N^{-1} endpoint layer; balancing the first two terms in Equation (S299) gives $1 - \alpha \asymp (NJ)^{-1/3}$ and the rate in Equation (S300). When $J \gg N^2$, expansion of the finite-sum derivative at one instead gives $1 - \alpha \asymp N/J$; the improvement over the DMOP-1 envelope is then of smaller order than its leading N^2/J term. These are comparisons of the conditional upper envelope, not strict orderings of the actual MSE. Such an ordering would require the signed endpoint conditions in Equations (S303)–(S304).

Remark 5 (What is known at the endpoints). Theorem S6 supplies only upper bounds. There is no DMOP-0 MSE lower bound in the present assumptions: the conditional population biases can cancel, and in special models the population bias can vanish. Therefore the displayed upper bounds cannot prove that an interior choice of α has smaller MSE than DMOP-0.

For every fixed N , Theorem S6 gives the ordinary Monte Carlo rate $O(J^{-1})$ at $\alpha = 1$; the term N^4/J^2 is then of smaller order as $J \rightarrow \infty$. A rate slower than J^{-1} appears only in a joint tuning argument in which $\alpha = \alpha_J < 1$ is sent toward one while J grows. In that argument, the nonzero population discount bias is reduced at the price of lengthening the particle genealogy. The unconditional bound proved here has variance term N^3/J independent of α , so it does not itself justify an interior choice. The conditional $J^{-2/3}$ MSE in Equation (S300) describes what would follow from the additional centered two-lineage estimate; it is not a deterioration of the fixed- N Monte Carlo rate.

Theorem 1 of Poyiadjis et al. (2011) concerns their standard path-space particle estimator of an unnormalized additive functional $\sum_{n=1}^N \varphi(X_n)$. Under their mixing and nondegeneracy conditions, the asymptotic variance in the $J^{1/2}$ central limit theorem is bounded below by $c_2 N^2 + c_1 N + c_0$, with $c_2 > 0$. Thus the corresponding central limit theorem is a fixed- N , $J \rightarrow \infty$ result whose asymptotic variance coefficient grows at least quadratically when N is subsequently increased. In this iterated sense, the leading variance of the unscaled particle error corresponding to this fixed- N central limit theorem is of order at least N^2/J . They note that the quadratic growth persists when the latent process is independent and identically distributed. In contrast, the generic L^p path-space upper bound cited by Poyiadjis et al. (2011) is $O(N^2/\sqrt{J})$ in error, or $O(N^4/J)$ after squaring. The quartic quantity is therefore a generic upper bound that is not established to be sharp, rather than the Poyiadjis rate.

The pathwise reparameterization marks in DMOP-1 are not generally identical particle by particle to the complete-data likelihood-ratio marks of Poyiadjis et al. (2011), which include initial- and transition-density score terms. Equality of their population score targets does not identify their particle variances. Consequently, the Poyiadjis lower bound does not by itself give a DMOP-1 lower bound. Such a result would require either a particlewise identification under additional model conditions or a direct nondegenerate central limit theorem for the DMOP-1 marks. Theorem S2 gives fixed- N consistency as $J \rightarrow \infty$, but no joint N, J rate.

Remark 6 (Oscillation normalization and the path-space factor). Suppose a current-state particle theorem gives

$$\|\eta_N^J(h) - \eta_N(h)\|_{L^2} \leq \frac{C \operatorname{osc}(h)}{\sqrt{J}}. \quad (\text{S310})$$

For any test function h_N with $\text{osc}(h_N) = O(N)$, homogeneity gives RMSE $O(N/\sqrt{J})$ and MSE $O(N^2/J)$. Thus removing an $\text{osc}(h) \leq 1$ normalization contributes one factor N^2 to the MSE, not two.

A historical path-space theorem can contain a separate horizon factor. The occupation-measure estimate used in Equation (S235) gives

$$\text{RMSE}(\hat{H}_N) \leq C \sqrt{\frac{N}{J}} \text{osc}(H_N). \quad (\text{S311})$$

Substituting $\text{osc}(H_N) = O(N)$ gives RMSE $O(N^{3/2}/\sqrt{J})$ and MSE $O(N^3/J)$. The factor N^2 in MSE comes from the scale of the unnormalized additive score, and the remaining factor N comes from the generic historical occupation variance. A still coarser path-space inequality with an N , rather than $\sqrt{N}$, error factor would give RMSE $O(N^2/\sqrt{J})$ and MSE $O(N^4/J)$, but that quartic rate is not the bound used in Theorem S6. The quadratic asymptotic-variance lower bound of Poyiadjis et al. (2011) is already for an unnormalized additive functional. It must not be multiplied by a second normalization factor, and it does not assert that the quartic upper bound is sharp.

S8 Figures for Bayesian Inference

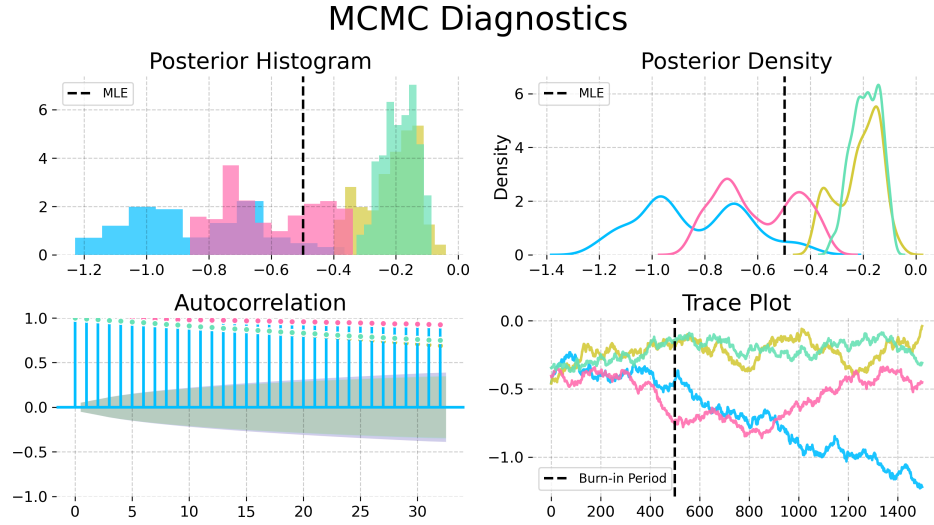


Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

MCMC Diagnostics

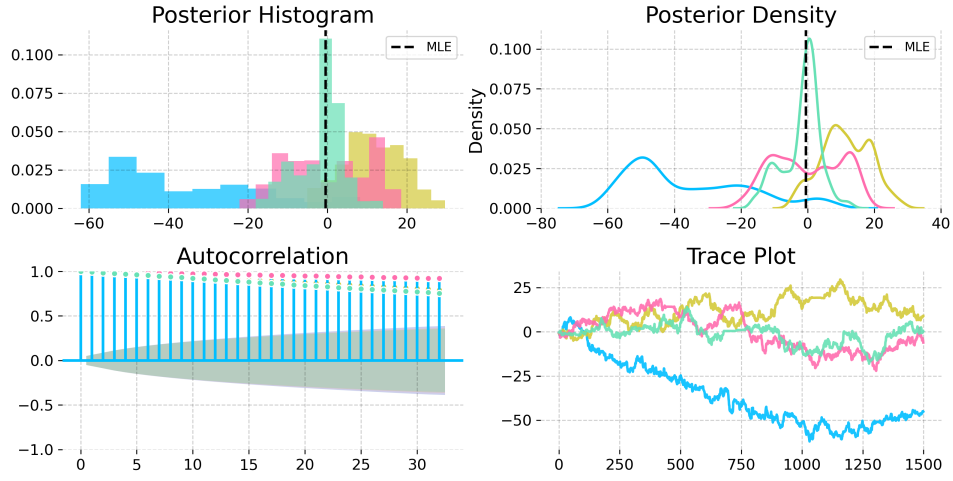


Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

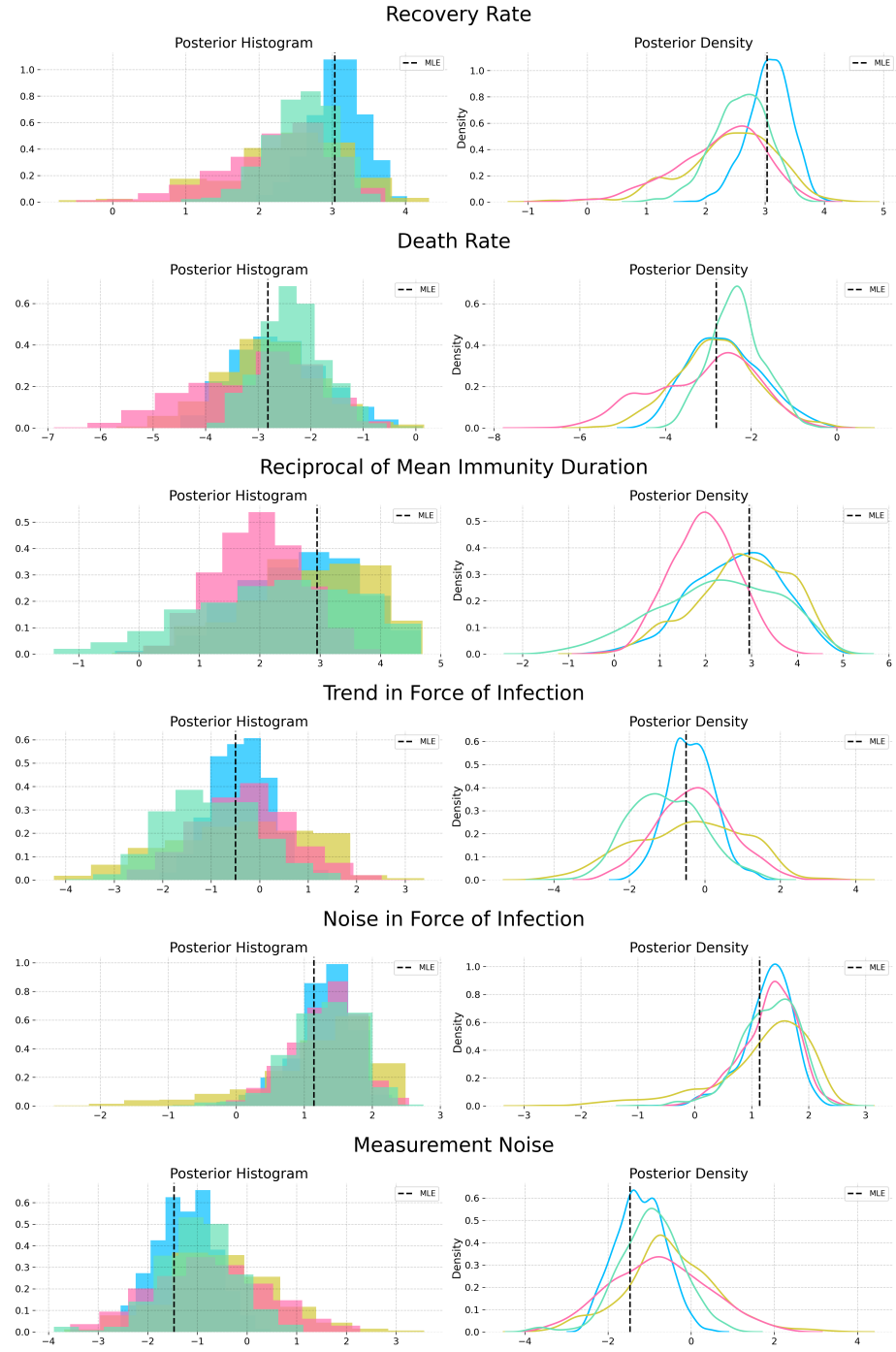


Figure S-3: Posterior estimates from NUTS powered by MOP- α with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016), 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as

the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0 , are treated differently: they are perturbed only at time t_0 , with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	Standard IF2	IFAD-0	IFAD-0.97	IFAD-1
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	650	60	175	60
GD Iterations	—	225	175	225
IF2 Cooling Rate	0.8	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemented by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not require it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

Supplementary References

- Andrieu, C., Doucet, A., and Holenstein, R. (2010). Particle Markov chain Monte Carlo methods. *Journal of the Royal Statistical Society, Series B*, 72:269–342.
- Bakker, K. M., Martinez-Bakker, M. E., Helm, B., and Stevenson, T. J. (2016). Digital epidemiology reveals global childhood disease seasonality and the effects of immunization. *Proceedings of the National Academy of Sciences*, 113(24):6689–6694.
- Becker, A. D., Birger, R. B., Teillant, A., Gastanaduy, P. A., Wallace, G. S., and Grenfell, B. T. (2016). Estimating enhanced prevaccination measles transmission hotspots in the context of cross-scale dynamics. *Proceedings of the National Academy of Sciences*, 113(51):14595–14600.
- Blackwood, J. C., Cummings, D. A. T., Broutin, H., Iamsirithaworn, S., and Rohani, P. (2013a). Deciphering the impacts of vaccination and immunity on pertussis epidemiology in Thailand. *Proceedings of the National Academy of Sciences*, 110:9595–9600.
- Blackwood, J. C., Streicker, D. G., Altizer, S., and Rohani, P. (2013b). Resolving the roles of immunity, pathogenesis, and immigration for rabies persistence in vampire bats. *Proceedings of the National Academy of Sciences*.
- Blake, I. M., Martin, R., Goel, A., Khetsuriani, N., Everts, J., Wolff, C., Wassilak, S., Aylward, R. B., and Grassly, N. C. (2014). The role of older children and adults in wild poliovirus transmission. *Proceedings of the National Academy of Sciences*, 111(29):10604–10609.
- Briga, M., Goult, E., Brett, T. S., Rohani, P., and Domenech de Cellès, M. (2024). Maternal pertussis immunization and the blunting of routine vaccine effectiveness: A meta-analysis and modeling study. *Nature Communications*, 15(1):921.
- Calafat, F. M., Wahl, T., Lindsten, F., Williams, J., and Frajka-Williams, E. (2018). Coherent modulation of the sea-level annual cycle in the United States by Atlantic Rossby waves. *Nature Communications*, 9(1):2571.
- Chan, H. P., Del Moral, P., and Jasra, A. (2018). A sharp first order analysis of Feynman–Kac particle models. *Stochastic Processes and their Applications*, 128(1):332–353.
- Chen, Y., Gerber, M., Andrieu, C., and Douc, R. (2025). Self-organizing state-space models with artificial dynamics. *Journal of the Royal Statistical Society, Series B*, page qkaf069.
- Chopin, N. (2004). Central limit theorem for sequential Monte Carlo methods and its application to Bayesian inference. *Annals of Statistics*, 32:2385–2411.
- Chopin, N. and Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*. Springer.
- Del Moral, P. (2004). *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, New York.
- Del Moral, P. and Doucet, A. (2003). On a class of genealogical and interacting Metropolis models. In Azéma, J., Émery, M., Ledoux, M., and Yor, M., editors, *Séminaire de Probabilités XXXVII*, pages 415–446. Springer, Berlin.

- Del Moral, P. and Rio, E. (2011). Concentration inequalities for mean field particle models. *The Annals of Applied Probability*, 21(3).
- Donoho, D. (2017). 50 years of data science. *Journal of Computational and Graphical Statistics*, 26(4):745–766.
- Fasiolo, M., Pya, N., and Wood, S. N. (2016). A comparison of inferential methods for highly nonlinear state space models in ecology and epidemiology. *Statistical Science*, 31(1):96–118.
- Fox, S. J., Lachmann, M., Tec, M., Pasco, R., Woody, S., Du, Z., Wang, X., Ingle, T. A., Javan, E., Dahan, M., Gaither, K., Escott, M. E., Adler, S. I., Johnston, S. C., Scott, J. G., and Meyers, L. A. (2022). Real-time pandemic surveillance using hospital admissions and mobility data. *Proceedings of the National Academy of Sciences*, 119(7):e2111870119.
- Giezendanner, J., Pasetto, D., Perez-Saez, J., Cerrato, C., Viterbi, R., Terzago, S., Palazzi, E., and Rinaldo, A. (2020). Earth and field observations underpin metapopulation dynamics in complex landscapes: Near-term study on carabids. *Proceedings of the National Academy of Sciences*, 117(23):12877–12884.
- Gu, M. G. and Kong, F. H. (1998). A stochastic approximation algorithm with Markov chain Monte-Carlo method for incomplete data estimation problems. *Proceedings of the National Academy of Sciences*, 95(13):7270–7274.
- Hao, Y., Abkemeier, A. J., and Ionides, E. L. (2024). Poisson approximate likelihood compared to the particle filter. *arXiv:2409.12173*.
- He, D., Lin, L., Artzy-Randrup, Y., Demirhan, H., Cowling, B. J., and Stone, L. (2023). Resolving the enigma of Iquitos and Manaus: A modeling analysis of multiple COVID-19 epidemic waves in two Amazonian cities. *Proceedings of the National Academy of Sciences*, 120(10):e2211422120.
- He, K., Hao, Y., and Ionides, E. L. (2025). Poisson Approximate Likelihood versus the block particle filter for a spatiotemporal measles model. *arXiv:2507.09121*.
- Ionides, E. L., Bretó, C., and King, A. A. (2006). Inference for nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 103:18438–18443.
- Ionides, E. L., Nguyen, D., Atchadé, Y., Stoev, S., and King, A. A. (2015). Inference for dynamic and latent variable models via iterated, perturbed Bayes maps. *Proceedings of the National Academy of Sciences*, 112(3):719–724.
- Karjalainen, J., Lee, A., Singh, S. S., and Vihola, M. (2023). On the forgetting of particle filters. *arXiv:2309.08517*.
- King, A. A., Ionides, E. L., Pascual, M., and Bouma, M. J. (2008). Inapparent infections and cholera dynamics. *Nature*, 454:877–880.
- Kingma, D. P. and Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv:1412.6980*.
- Laneri, K., Paul, R. E., Tall, A., Faye, J., Diene-Sarr, F., Sokhna, C., Trape, J.-F., and Rodó, X. (2015). Dynamical malaria models reveal how immunity buffers effect of climate variability. *Proceedings of the National Academy of Sciences*, 112(28):8786–8791.

- Li, R., Pei, S., Chen, B., Song, Y., Zhang, T., Yang, W., and Shaman, J. (2020). Substantial undocumented infection facilitates the rapid dissemination of novel coronavirus (SARS-CoV-2). *Science*, 368(6490):489–493.
- Lindsten, F., Jordan, M. I., and Schön, T. B. (2014). Particle Gibbs with ancestor sampling. *The Journal of Machine Learning Research*, 15(1):2145–2184.
- Martinez, P. P., King, A. A., Yunus, M., Faruque, A., and Pascual, M. (2016). Differential and enhanced response to climate forcing in diarrheal disease due to rotavirus across a megacity of the developing world. *Proceedings of the National Academy of Sciences*, 113(15):4092–4097.
- Naesseth, C., Linderman, S., Ranganath, R., and Blei, D. (2018). Variational sequential Monte Carlo. In Storkey, A. and Perez-Cruz, F., editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 968–977. PMLR.
- Pei, S., Liljeros, F., and Shaman, J. (2021). Identifying asymptomatic spreaders of antimicrobial-resistant pathogens in hospital settings. *Proceedings of the National Academy of Sciences*, 118(37):e2111190118.
- Perez-Saez, J., Bellon, M., Lessler, J., Berthelot, J., Hodcroft, E. B., Michielin, G., Pennacchio, F., Lamour, J., Laubscher, F., L’Huillier, A. G., Posfay-Barbe, K. M., Maerkl, S. J., Guessous, I., Azman, A. S., Eckerle, I., Stringhini, S., and Lortie, E. (2025). Evolving infectious disease dynamics shape school-based intervention effectiveness. *Nature Communications*, 16(1):6597.
- Pons-Salort, M. and Grassly, N. C. (2018). Serotype-specific immunity explains the incidence of diseases caused by human enteroviruses. *Science*, 361(6404):800–803.
- Poyiadjis, G., Doucet, A., and Singh, S. S. (2011). Particle approximations of the score and observed information matrix in state space models with application to parameter estimation. *Biometrika*, 98(1):65–80.
- Ranjeva, S., Subramanian, R., Fang, V. J., Leung, G. M., Ip, D. K., Perera, R. A., Peiris, J. M., Cowling, B. J., and Cobey, S. (2019). Age-specific differences in the dynamics of protective immunity to influenza. *Nature Communications*, 10(1):1660.
- Ranjeva, S. L., Baskerville, E. B., Dukic, V., Villa, L. L., Lazcano-Ponce, E., Giuliano, A. R., Dwyer, G., and Cobey, S. (2017). Recurring infection with ecologically distinct HPV types can explain high prevalence and diversity. *Proceedings of the National Academy of Sciences*, 114(51):13573–13578.
- Rodó, X., Martinez, P. P., Siraj, A., and Pascual, M. (2021). Malaria trends in Ethiopian highlands track the 2000 ‘slowdown’ in global warming. *Nature Communications*, 12(1):1555.
- Roosta-Khorasani, F. and Mahoney, M. W. (2016). Sub-sampled Newton methods I: Globally convergent algorithms. *arXiv:1601.04737*.
- Santos-Vega, M., Martinez, P., Vaishnav, K., Kohli, V., Desai, V., Bouma, M., and Pascual, M. (2022). The neglected role of relative humidity in the interannual variability of urban malaria in Indian cities. *Nature Communications*, 13(1):533.

- Ścibior, A. and Wood, F. (2021). Differentiable particle filtering without modifying the forward pass. *arXiv:2106.10314*.
- Subramanian, R., He, Q., and Pascual, M. (2021). Quantifying asymptomatic infection and transmission of COVID-19 in New York City using observed cases, serology, and testing capacity. *Proceedings of the National Academy of Sciences*, 118(9).
- Wale, N., Jones, M. J., Sim, D. G., Read, A. F., and King, A. A. (2019). The contribution of host cell-directed vs. parasite-directed immunity to the disease and dynamics of malaria infections. *Proceedings of the National Academy of Sciences*, 116(44):22386–22392.
- Wang, E. T., Vannucci, M., Haneef, Z., Moss, R., Rao, V. R., and Chiang, S. (2022). A Bayesian switching linear dynamical system for estimating seizure chronotypes. *Proceedings of the National Academy of Sciences*, 119(46):e2200822119.
- Whitehouse, M., Whiteley, N., and Rimella, L. (2023). Consistent and fast inference in compartmental models of epidemics using Poisson Approximate Likelihoods. *Journal of the Royal Statistical Society, Series B*, 85(4):1173–1203.
- Whitehouse, M., Whiteley, N., and Rimella, L. (2025). Correction to: Consistent and fast inference in compartmental models of epidemics using Poisson approximate likelihoods. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, pre-published online.
- Wycoff, N., Smith, J. W., Booth, A. S., and Gramacy, R. B. (2026). Voronoi candidates for Bayesian optimization. *Journal of Global Optimization*, 94:175–201.